



THE UNIVERSITY OF ZAMBIA
INSTITUTE OF DISTANCE EDUCATION

BACHELOR OF ARTS
(Economics)

MODULE

INTRODUCTION TO MACROECONOMIC THEORY
(ECN 1215)

March, 2014

© COPYRIGHT

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording or otherwise without the permission of the University of Zambia, Institute of Distance Education.

Inquiries concerning reproduction or rights and requests for additional training materials should be addressed to:

The Director,

Institute of Distance Education

The University of Zambia

P.O. Box 32379

Lusaka

Zambia

Tel: +211 290719

Fax: +211 253952

E-mail: director-ide@unza.zm

Website: www.unza.zm

SERVICE AND EXCELLENCE

ACKNOWLEDGEMENT

The University of Zambia (UNZA), Institute of Distance Education (IDE) wishes to thank Obrian Ndhlovu for writing this module.

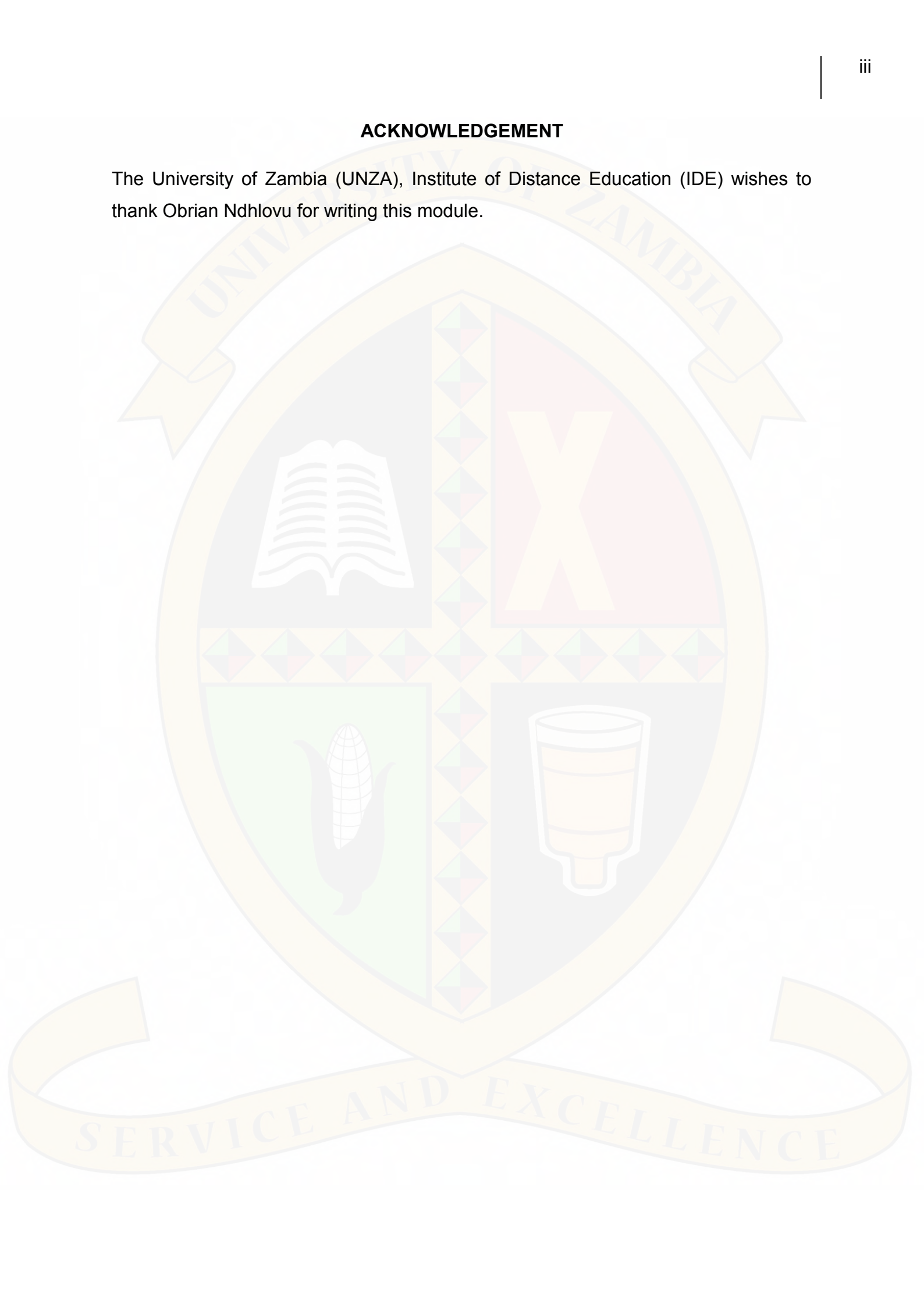


TABLE OF CONTENTS

Module overview	viii
Introduction	viii
Module Objectives	viii
Timeframe	viii
Study skills.....	viii
Need help?	ix
Assessments	x
Required Reading.....	x
Recommended Reading	x
1 Study and Methodology of Macroeconomics	1
1.1 Introduction	1
1.2 Contents list	1
1.3 Statement of pre – requisite knowledge	1
1.4 Learning objectives for the unit	2
1.5 Supplementary Reading.....	2
1.6 Time required	2
1.7 Basic Definitions.....	2
1.8 Normative and Positive approach: Policy relevance	3
1.9 Unit Summary	7
2 National Income Accounts	8
2.1 Introduction	8
2.2 Content List.....	8
2.3 Statement of pre – requisite knowledge	8
2.4 Learning objectives for the unit	9
2.5 Supplementary Reading.....	9
2.6 Time required	9
2.7 Circular Flow	9
2.7.1 The Simple Economy without Government.....	9
2.7.2 The simple Economy with Government	12
2.7.3 An Open Economy	13

2.8	National Income Accounting.....	14
2.9	Inadequacies of GDP as a measure of Economic Activity	16
2.10	Real vs. nominal – price indices	17
2.11	Determinants of national income; income and expenditure model.....	19
2.12	Unit Summary	21
3	Unemployment.....	22
3.1	Introduction	22
3.2	Contents list	22
3.3	Statement of pre – requisite knowledge	22
3.4	Learning objectives for the unit	22
3.5	Supplementary Reading.....	23
3.6	Time Required.....	23
3.7	Definition	23
3.8	Types and Causes of Unemployment	25
3.9	Effects of Unemployment	29
3.10	Cures of Unemployment	29
3.11	Unit Summary	31
4	Government Sector	32
4.1	Introduction	32
4.2	Contents list	32
4.3	Statement of pre – requisite knowledge	33
4.4	Learning objectives for the unit	33
4.5	Supplementary Reading.....	33
4.6	Time required	33
4.7	Public Finance.....	33
4.7.1	Taxes.....	34
4.7.2	Retrogressive, Proportional and Progressive Taxes.....	35
4.8	Budget including Debt Management	36
4.8.1	Balanced Budget and Debt Management	37
4.9	Fiscal Policy	38
4.10	Unit Summary	38
5	Monetary Sector	40
5.1	Introduction	40

5.2	Contents list	40
5.3	Statement of pre – requisite knowledge	40
5.4	Learning objectives for the unit	40
5.5	Supplementary Reading.....	41
5.6	Time required	41
5.7	Money	41
5.7.1	Functions of Money	41
5.7.2	Characteristics of Money	42
5.8	The banking system and the central bank.....	46
5.8.1	The Banking System.	46
5.8.2	The Central Bank.....	47
5.9	Monetary policy	48
5.9.1	Instruments of Monetary Policy	51
5.10	Unit Summary	53
6	Inflation	54
6.1	Introduction	54
6.2	Contents list	54
6.3	Statement of pre – requisite knowledge	54
6.4	Learning objectives for the unit	54
6.5	Supplementary Reading.....	55
6.6	Time required	55
6.7	Definitions	55
6.8	Causes.....	55
6.8.1	The Monetarist View	56
6.8.2	The Keynesian.....	57
6.8.3	Imported Inflation.....	60
6.9	Effects of Inflation.....	60
6.9.1	Anticipated Inflation	60
6.9.2	Unanticipated Inflation	61
6.10	Cures for Inflation.....	63
6.11	Unit Summary	65
7	External Sector.....	66
7.1	Introduction	66

7.2	Contents list	66
7.3	Statement of pre – requisite knowledge	66
7.4	Learning objectives for the unit	66
7.5	Supplementary Reading.....	67
7.6	Time required	67
7.7	Trade.....	67
7.8	A Justification for Trade	68
7.8.1	Absolute Advantage	68
7.8.2	Comparative Advantage	70
7.9	Balance of Payment	72
7.9.1	Current Account.....	72
7.9.2	Capital Account	74
7.10	Exchange Rates.....	76
7.11	Unit Summary	78
8	Module Summary	79

MODULE OVERVIEW

Introduction

This module covers the course „introduction to Macroeconomics Theory“, an introductory level course to the study of economics. It is intended to introduce students to the macro-part of economics. In the series, it is a precursor to Intermediate Macroeconomics Theory which will build up on almost all concepts developed in this module.

Module Objectives

By the end of this course students are expected to:

- Demonstrate an understanding of the nature and methodology of macroeconomics.
- Show knowledge of the concepts of national income determination, money and banking, public finance, and international trade.
- Demonstrate the foundation necessary for the study of intermediate macroeconomics.

Timeframe

This module must be covered within 14 weeks.

Study skills

You may not have studied by distance education before. Here are some tips that will help you do better in your learning and keep you focused-

1. Set goals such as: I will succeed in this course. At the beginning of the module, break the lessons into manageable chunks. You might not have time to do a full lesson in one night, so plan how much you can do, then stick to it until you are done.
2. Establish a regular study/learning schedule
3. Determine what time is best for you to study
4. Have a dedicated study place with all the supplies you might need

5. Tell people what you are doing because only then are you more likely to stick to a course.
6. Ask someone to proofread your work before you submit it.
7. If you do not understand something ask your local learning centre or your tutor, who will be able to help you.
8. Search for the meaning of principles and concepts instead of just memorizing

Need help?

In case you have difficulties during the duration of the course, please get in touch with the Director, Institute of Distance Education, or the resident lecturer in your province.

All enquiries in connection with the payment of fees should be directed to the Director, Institute of Distance Education:

The Director,

Institute of Distance Education,
University of Zambia,
P. O. Box 32379,
10101 Lusaka

Coordinator, Learner Support Services (Land Cell): +260 978772248

Senior Administrative Officer

(Programme Development & Production) +260 977639993

IDE Land Line: +260 211 290719

IDE Fax: +260 211 290719

IDE E-mail: director-ide@unza.zm

<http://www.unza.zm>

SERVICE AND EXCELLENCE

Assessments

Continuous Assessment	50 %
One assignment	20%
Two tests	30%
Final Examination	50 %

Required Reading

Begg, D., S. Fischer and R. Dornbusch (2003). Economics. (7th Edition). London: MaGraw-Hill.

Recommended Reading

Hardwick, P. B. Khan and J. Langmead (2006). Introduction to Modern Economics. London: Longman, 2006.

UNIT 1

1 STUDY AND METHODOLOGY OF MACROECONOMICS

1.1 Introduction

The study of Economics theory is divided into two, micro and macro. As the names might suggest, the former is concerned with micro components of the economy. It looks at the economics of entities whose behaviour does not affect the whole economy. The latter on the other hand is more concerned with the macro part, the aggregate of the economy. It looks at fundamentals like unemployment, inflation rate, government expenditure and taxation. As rightly put by Snowdon and Vane (2005, p1).

“Macroeconomics is concerned with the structure, performance and behaviour of the economy as a whole. The prime concern of macroeconomists is to analyse and attempt to understand the underlying determinants of the main aggregate trends in the economy with respect to the total output of goods and services (GDP), unemployment, inflation and international transactions.”

This citation brings to light four main themes of macroeconomics. That is output, unemployment, inflation and international trade.

1.2 Contents list

- Basic Definitions
- Normative and positive approach; policy relevance.

1.3 Statement of pre – requisite knowledge

None.

1.4 Learning objectives for the unit

By the end of this unit, students are expected to demonstrate an understanding of the nature and methodology of macroeconomics

1.5 Supplementary Reading

Begg, D., S. Fischer and R. Dornbusch (2003) Economics. (7th Edition). London: McGraw-Hill

Hardwick, P., B. Khan and J. Langmead (2006) Introduction to Modern Economics. London: Longman

Dornbusch, R. Et al (2004) Macroeconomics 9th ed.

Langmead, P et al. Tax Policy issues in Zambia: Selected Papers, Lusaka

1.6 Time required

At most one week

1.7 Basic Definitions

The basic understanding in economics is that there is an imbalance between what is wanted and what is actually available. The sum of all wants in the economy far exceeds what can be provided in the economy. This is the bedrock of economics, a science of choice because there isn't enough. Thus, economics can be defined as the study of allocation of *scarce* resources among *alternative* uses. The definition brings out two important terms in economics; scarcity and alternative use.

The term scarcity is used in economics to mean the inadequacy of resources such as land, labour and capital. These are the resources that go into producing anything that man wants. The wants on the other hand tend to be unlimited, they are uncountable. Because the wants are not fully supported by the available means, choices have to be made on which wants to get. A requisite for choice is that the

resource must be flexible, to be used in alternative ways. In the absence of this flexibility, choice would not be possible.

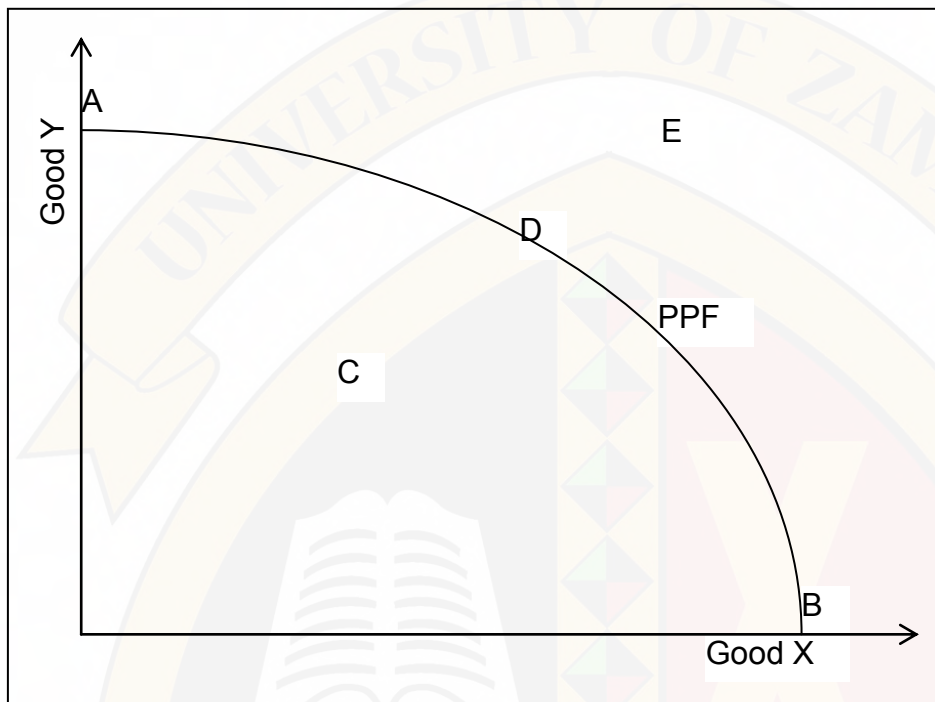
1.8 Normative and Positive approach: Policy relevance

Economics often involves making some judgment on the performance of the economy. Often we hear statements such as, „Government has created 2000 jobs“, or that „the economy grew by 6 percent between 2009 and 2010. These statements are stating facts which can be measured and verified. If government says it has created so many jobs, we can ask for a tabulation of the jobs created and prove the information. The same applies for the growth rate. All we need is the GDP levels for the two year and a calculator on the side.

Another set of statements are those that do not state facts. For instance, the statement, „government has the potential to create 600 jobs“, or the „Zambia’s economy can grow at above 10 percent if well managed“. These statements are not stating facts. As such, they are very difficult to verify.

In the study of economics, we distinguish between two approaches to looking at the economy. These are Normative and Positive approach. A normative approach is based on the „ought to be“ kind of assessment. It looks at the economy the way it ought to be, the way it is expected to be. It is associated with statements not stating facts, but makes value judgments about the economy. The popular concept is the concept of the *Production Possibility Frontier* (PPF). The PPF shows the maximum output in a two goods economy when all resources are fully and efficiently employed using the best technology.

Figure 1.1. PPF for X and Y



Assume an economy produces only two goods; X and Y. When all the conditions of full and efficient employment are reached, output will fall on the PPF. See *Figure 1.1 below*. The country can decide on any of the infinite number of points on the PPF. The slope of the PPF is called the *Marginal Rate of Transformation* (MRT).

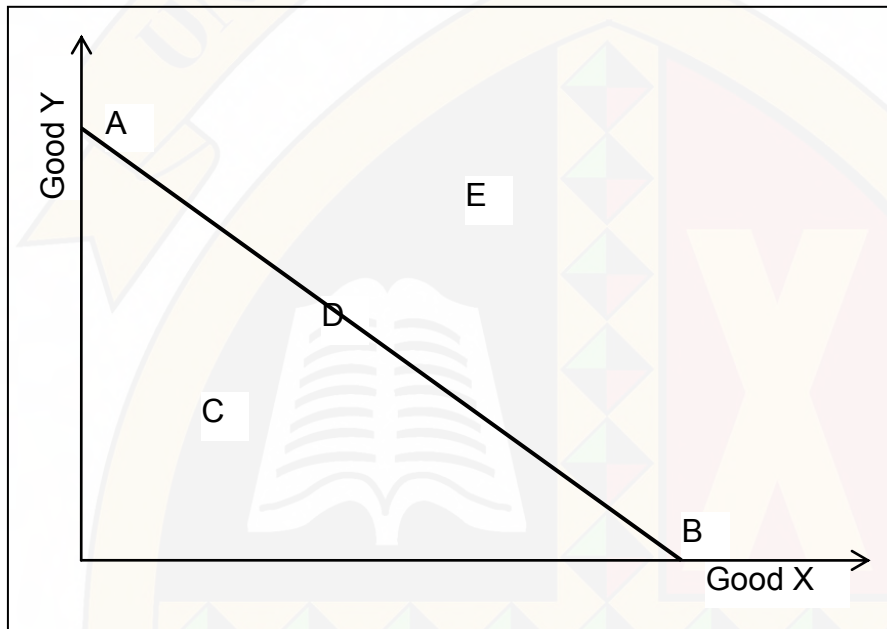
$$MRT = \frac{\Delta y}{\Delta x}$$

It defines the rate at which the production of one good (measured in the vertical axis) can be transformed into the production of another by way of reallocation of productive resources. The MRT shows the loss in the output of good Y that is made in order to produce one more unit of the good X. In other words, the MRT shows the marginal opportunity cost of producing good X given in terms of good Y. To produce an additional unit of X, we have to forgo so much of Y. From its name, it shows the rate at which one product is transformed into another.

In reality, the MRT will increase due to increasing marginal “opportunity” cost of producing X as more of it is produced. This is the reason the PPF always takes a concave shape from the origin. It is flatter for lower levels of X and much steeper for

higher levels of X . In order to simplify the analysis, it is possible to assume a constant MRT. This entails a linear PPF. Every additional unit of X requires forgoing the same amount of Y . This is shown in Figure 1.2 below.

Figure 1.2. PPF of X and Y with Constant MRT



Now let us consider the two figure, Figure 1.1 and Figure 1.2. Both have been drawn with similar points in corresponding positions. If the country only produces one commodity, X , it will be at point B and if producing Y only, then point A applies. Naturally, it will produce both goods and will be anywhere between the two extreme points such as point D. Two other points, C and E fall off the PPF.

Point C falls inside or below the PPF. A country would produce at point C if at least one of the conditions for the PPF is not fulfilled. That is, when the country is not at full employment or there is some inefficiency in the use of resource. It could also be the case of both underemployment and inefficiency. A country producing at such a point still has room for improvement. It can improve either by being efficient or employing all the resources or both.

Point E is outside the PPF. Since the PPF represents what can be produced when everything is used in the most efficient way, it means any output outside the PPF is not tenable. Thus, point E is outside the possibility, it is not possible for the economy to reach that level given the same resources and technology.

We must be very clear on the implication of point E here. It is impossible today, with the current state of affairs. It does not however mean the country will never reach there. Really! First the technology keeps advancing. Second, the resources available to a country such as labour, capital are also not constant. When a change in any of these factors occurs, the PPF shifts (tilts in the case of commodity specific discoveries) outward. Point E will become tenable when the shift in the PPF causes E to fall within.

Positive economics on the other hand is based on facts. It is an approach to economics which looks at the economy the way it is without attaching value or normative judgments. It concerns itself with „as is“. Using the same PPF in Figure 1.1 and assuming an economy producing below the PPF at point C. Positive economics would simply report that the economy is performing at this level. It produces so much of one commodity and so much of the other. In contrast to this, the normative economics will look at the potential, that the economy has the potential to produce at point D which is higher than C.

The distinction between Normative and Positive economics is vital in macroeconomics. While positive economics will report the situation as it is, the normative economics by reporting on the „ought to be“ informs policy direction. Since policies are formulated in order to move the economy to its full potential, the gap between the actual level (positive economics) and the potential level (normative economics) will give direction to type and nature of policy that is needed.

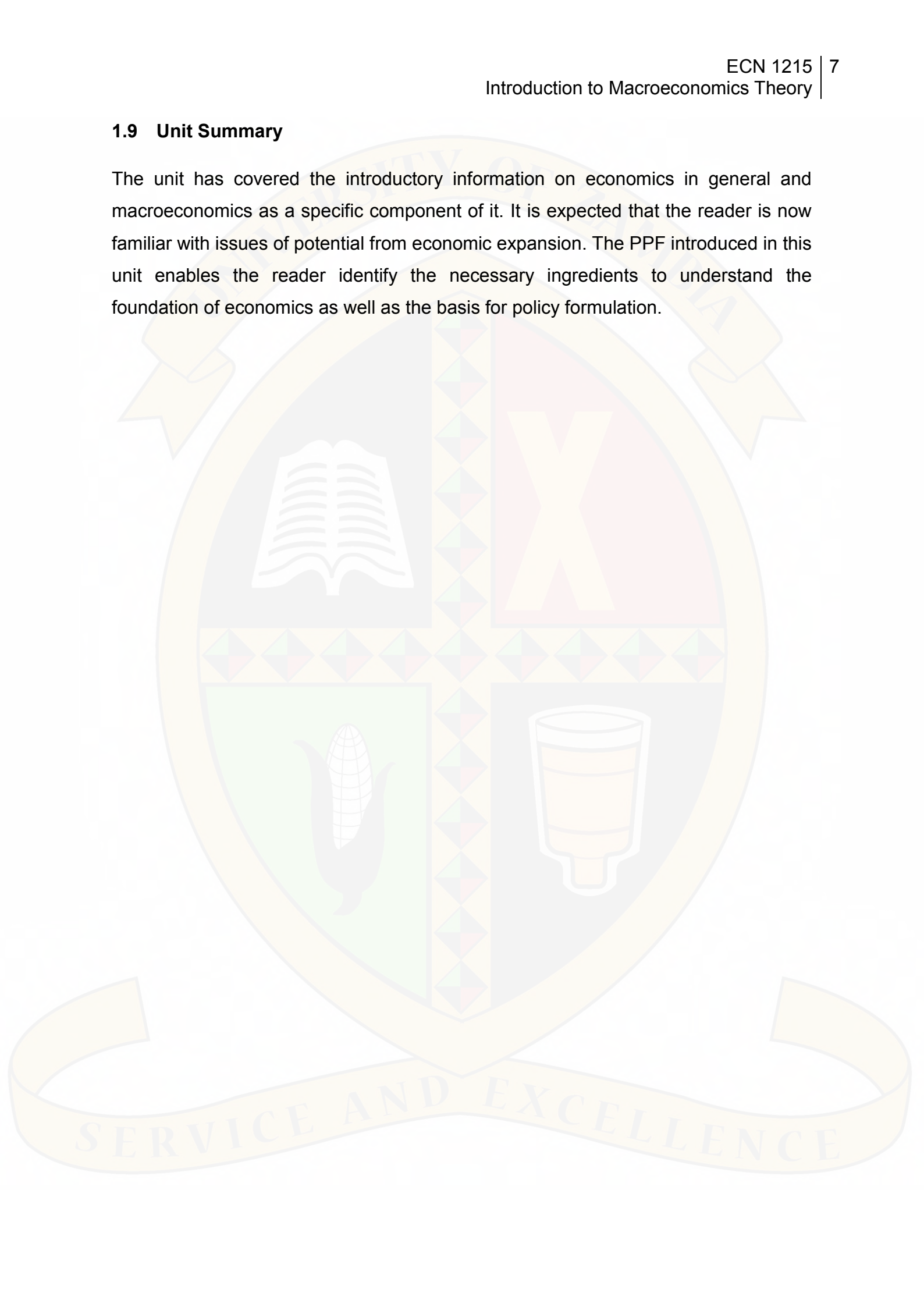
Activity 1-1

Country A only produces two good, industrial output X and mining output Y. Use a well labelled PPF to show the following changes.

- i. An improvement in mining technology which allows for increased mining output.*
- ii. An increase in population*

1.9 Unit Summary

The unit has covered the introductory information on economics in general and macroeconomics as a specific component of it. It is expected that the reader is now familiar with issues of potential from economic expansion. The PPF introduced in this unit enables the reader identify the necessary ingredients to understand the foundation of economics as well as the basis for policy formulation.



UNIT 2

2 NATIONAL INCOME ACCOUNTS

2.1 Introduction

During the course of your life or career, you probably have already encountered or taken part in a transaction. You have sold or bought some good or service before. If you are working or have worked before, you are selling your labour and your employer buys by paying a price which we will refer to as a wage (or salary). An entrepreneur receives profit while an investor receives interest. All this is income and can be measured in different ways.

This unit is intended to bring out measurement issues in economics. It primarily looks at the best way of measuring economic activity. If there is growth in the economy, what exactly is meant by that? This unit will tackle the three different ways of measuring economic activity. These are Income method, Expenditure method and the Product method.

2.2 Content List

- Circular Flow.
- National Income Accounting
- Inadequacies of GDP as a measure of Economic Activity
- Real vs. nominal variables – price indices
- Determinants of national income; income and expenditure model

2.3 Statement of pre – requisite knowledge

This unit does not assume any pre-knowledge on economics. You are nonetheless expected to have some basic knowledge of elementary mathematics.

2.4 Learning objectives for the unit

By the end of this unit, you should be able to demonstrate knowledge of the concepts of national income determination.

2.5 Supplementary Reading

Begg, D., S. Fischer and R. Dornbusch (2003) Economics. (7th Edition). London: McGraw-Hill

Hardwick, P., B. Khan and J. Langmead (2006) Introduction to Modern Economics. London: Longman

Dornbusch R and S. Fischer (1993) Macroeconomics

Mankiw N.G. (2003) Macroeconomics

2.6 Time required

This unit must be covered in not more than two weeks.

2.7 Circular Flow

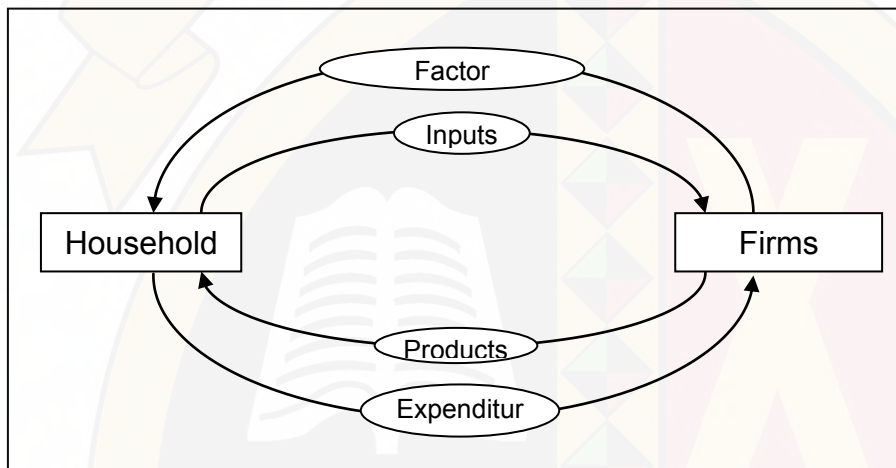
Many transactions take place in an economy. If we have to give an illustration of all these transaction, it would be so complex that making sense of it would be difficult. As a first step, we must envision a simple economy from where we can classify the many transactions with ease. We develop this in three stages. The first will be the simplest of all; the simple economy, comprising of households and firms only. The second will bring in the third component, the government. The third will be an open economy, one that has the external sector. Exports and imports take place under this model.

2.7.1 The Simple Economy without Government

Think of a simple economy with Households (HH) and Firms only. As individuals, we fall under households. Any enterprises or producing entities will fall under firms. Of course the real distinction may be obscure in some cases. For instance, a peasant farmer produces and consumes his own food. In this case, he is both a firm (because he is a producer) and a household (because he is a consumer).

In this particular case, economic theory dictates that the two components of production and consumption be delinked. This will give the *family* and the *farm*. The former owns the latter. In a more general case, the farm is a firm while the family is a household. The interaction of the two is demonstrated in what is called the Circular Flow of Income.

Figure 2.1: Circular Flow of Income for a closed economy without Government



In Figure 2.1 above, the four income paths show the transactions between the households and firms. The shape is in a circular form. This is the reason it is called *Circular Flow of Income*. It shows income flowing in a circular form between the household and firms. Now let us look at each of these income paths, from the top to the bottom.

For a firm to exist and produce, it requires factors of production. These are capital, labour and entrepreneurship. For this unit, these will collectively be referred to as inputs or factors. But these are owned by the households, the people. They supply these factors to the firms, not for free anyway. They sell these factors to firms. This is shown by the flow labelled as *inputs*.

In return, firms pay for the factors of production received from households. For the labour, the firm pay a salary or wage. For the capital, interest is paid to the owners. To the entrepreneur, a profit is paid by the firm. All these payments are made by the firm to the households in exchange for the factors of production they receive. Collectively, such payments are called *factor payments*. It is payment for the factors of production.

Since factor payment is made in exchange for the factors received, the payment is equivalent to the value of the factor offered. The total value of labour supplied must equal the total wages paid. The same applies to all the other factors of production. This will mean the two arrows are of equal value. We can calculate one and know that the other is exactly equal to that. This is on the factor market side.

The employment of labour and other complementing factors results into a product. This is the sole purpose of employing factor of production. It is to produce something, a good or a service. Specific examples include bread, maize, and transportation services. These, in their final form¹, are sold to households. This flow is represented by the flow labelled *product*. It summarises everything that firms produce. It can be a physical good such as bread, bottled water or a service such as cleaning services, transportation service.

The *product* flow however does not include any intermediate product. The dividing line might be thin. But strictly speaking, economics views any product produced by a firm and sold to another firm as an intermediate product. A good is said to be final if it is sold to the households. For instance, bread is a final good when sold to households. It is intermediate when bought by another firm for instance a hotel or lodge.

In exchange for the products that households receive from firms, they must transfer money to firms. This is really an expenditure on the products from the firms. This is represented by an *expenditure* flow. This coins all expenditure by the households on goods and services that firms produce. The total expenditure is equal to the value of goods and services produced. As such, the two flows represent flows of equal value.

The expenditure by the households on goods and services is funded by the incomes received when households supply productive resources to firms. Assuming there are no savings, the total income must equal total expenditure. All income that is received is spent. When the two equal, it means the four flows are equal. This completes the circular flow of income for a simple economy.

¹ There are two types of goods; intermediate and final. A good is intermediate if it is meant to be used in the production of another good. For instance, flour and cotton are all intermediate goods because they are meant for the production of bread and linen respectively. A final good on the other hand is one in consumable form. It is final and is sold to households. Therefore, bread and linen are final goods.

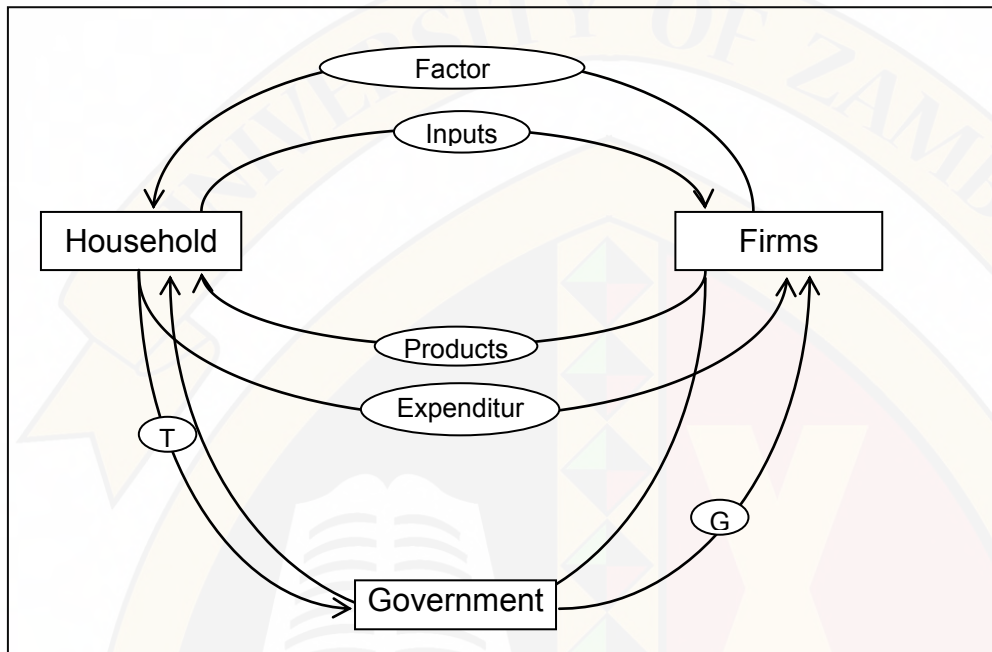
2.7.2 The simple Economy with Government

First ask yourself the following two questions. What does government do? And how does it differ from firms? Like firms, government produces goods and services. We all know that health and education services are world over predominantly produced by government. The situation is not different in Zambia. The education and health sectors are dominated by government. It also employs the different factors of production that households offer. It employs human resource. Nurse and Teachers in government institutions are not employed by firms like Lafarge Cement or Airtel Zambia. They are employed by government.

Government, like households also buys goods and services from firms. It will provide factors of production for the firms. Look at things like roads, or infrastructure in general. These are meant to aid firms in production. They are therefore factors of production because they are used in producing services and goods.

The difference however, is that government does not receive payments for what it provides. Government does not charge for a kilometre of a road used. Of course there are instances of direct payment for government services like school fees and a payment for a health centre visit. On the overall however, government services are funded by tax. Tax is imposed and paid by both households and firms. It need not be commensurate to the service received that a particular sector receives from government.

Figure 2.2: Circular flow of income for a simple economy with Government



With the inclusion of government, households are now not able to spend all their income. From the income received in the various names described earlier, part of it must be paid in taxes. Moreover, taxes are compulsory once imposed. It is the first thing households must clear before deciding on what else to buy. If you are working, the employer will just tell you that he has paid your taxes already and the remainder is what you get as *net pay*.

The taxes that government collects are used to pay for services that government gets from firms. For instance, government will not build the road itself, nor will it provide relief food. It will always hire firms to do that. The tax revenue, paid by households, is used to pay for such services which are part of output from firms. As such, total expenditure on output will now constitute what households spend as well as government expenditure on goods and services. We emphasise here that expenditure must be on goods and services. This is where government actually buys something.

2.7.3 An Open Economy

In an open economy, international trade is assumed to take place. The country exports part of its output. At the same time, consumption is not limited to domestically produced goods and services. Some are imported.

Activity 2-1

Draw the circular flow of income for an open economy.

2.8 National Income Accounting

In the previous subsection, you saw how income flows among the sectors of the economy. It was also shown that the different flows, in the case of a simple economy without government, actually represent flows of equal value. So it does not matter which flow you look at, you get the same value. Let us consider the flows, one at a time.

The factor payment flow. This flow includes all payments to factors of production. The actual list was given earlier in the unit. It represents the incomes to labour, capital owners etc though called by different technical names. Since we are summing for all factors of production in the domestic economy, the sum is referred to as Gross Domestic Income (GDI). You will see the acronym **GDI** used henceforth.

The GDI is the sum of all incomes received in the economy. It includes salaries, because they are paid in exchange for labour, interest because it is paid in exchange for capital rented to the firm. It also includes profits because they are paid for the entrepreneurship. It should not however include remittances such as government subsidies. Remittances do not represent any service or production. It just transfers money from one entity to another. In economics, they are collectively called **transfer payments**.

Activity 2-2

Government pays students at the University of Zambia Meal Allowances, colloquially referred to as BC. Should this be included in calculating GDI? Explain

The next is the product flow. This is a sum of all produce in the economy. Because this is being measured for a particular time period, usually a year, the measure is for produce produced within a particular year. It is what the country has produced in a

year, for instance 2013. Like the income flow, it is called the **Gross Domestic Product (GDP)**. That means summing the values of bread, copper, beef, etc produced in Zambia for that particular year. But something interesting here. Beef cannot be produced in a single year. Remember producing beef starts from the time a calf is conceived to the time its meat is put on a market. So how should this kind of production be accounted for?

Well, this case is not only limited to beef, many other products are produced over a span of several years. To measure such, value addition is used as opposed to total value. You probably have heard of **Value Added** in taxation. It is simply the value that has been added to a product in a particular year. Take for instance a calf born in the previous year. An estimate of its value is known as at 1st January. The farmer will do everything to make sure the calf results into beef. But by end of the year, though grown up, the animal is not ready yet to be slaughtered. The farmer closes the year without sells. Can we say he has not produced anything at all?

Absolutely no. Throughout the year, he was busy producing something but the product is just not ready yet. In this case, we resort to the value added in that particular year. It will be the difference between the value of value of the animal at the end of the year and the beginning. What was it worth at the beginning of the year? And what is it worth now as we close the year. The difference is the value added. It is the product for that particular year.

The third flow is the expenditure method. Like we have done for the first two, this is called **Gross Domestic Expenditure (GDE)**. It is a measure of all expenditure on goods and services in the domestic economy. It will sum all consumption by households, all government expenditure on goods and service as well as any expenditure by firms in investment. For an open economy, exports are part of domestic product, except they are sold outside the country. Thus they must be added. At the same time, the consumption which has been included may include goods not domestically produced. These are imported goods. They must be subtracted.

In summary, GDE is given by

$$GDE = C + I + G + X - M$$

the components on the right side are as follows. The C is household consumption, I is investment and G is the government expenditure on goods and services. The X and M represent exports and imports respectively.

So far, you have seen that GDP, the commonly used of the three, can actually be measured by looking at all incomes or all expenditures. The three measures are equal. The choice is a matter of convenience and availability of data. What about the fourth flow? The input flow is the sum of all factors of production used in production. However, owing to the complexity of tabulating the contribution of many factors of varying levels of productivities, this flow is not used in the measure of GDP. Suffice to say if measured, it would equal the GDP, GDI and GDE.

2.9 Inadequacies of GDP as a measure of Economic Activity

It is clear now that GDP or any other measure used is intended to measure the value of goods and services produced in an economy. The measure also represents the level of economic welfare of the residents (or nationals) of a country. The measures are however not free of shortcomings. Three specific major problems associated with GDP as a measure of economic activity or economic welfare. These are:

- I. **Poor measurement of Non-traded output:** GDP is a measure of the total value of all goods and services produced in the economy. This also equals the total incomes received. But there are certain activities that are not traded or bought from the market. Take a simple example of a house maid. The maid does all house chores and is paid an income. This is part of national income. It is part of national income because the maid has not received this as a gift but an earning. He/she will have worked for it. Now suppose instead of having a hired maid, a household has a relative doing the same work. Despite doing the same work, (equal output) the relative will not receive any income.
- II. **Does not capture value of reduced "bads":** GDP measurements are often restricted to tangible outputs or services. It looks at how much of various goods and services the economy has produced. It is blind to efforts or inputs made to fighting economic evils such as pollution and crime. Increased use of technology or industrialization often has a negative impact on the quality of natural resources such as land, atmosphere. Though the welfare of citizens or

residents will be better with low levels of crime or pollution, GDP measurement will see no value added to output. Thus, GDP fails to take a broader approach to measurement of welfare.

- III. Fails to account for improvements in quality: GDP measures the quantity of goods produced. Assume a car manufacturing economy. If 100 cars are manufactured every year, we say that there is change in GDP. This is precisely because the quantity has remained constant. Obviously, the economy may produce the same number of cars because they are investing more resources in the quality. Since better quality means higher value, this should result in increased GDP. One can still argue that this will be captured by increased price. But we bring out this point because of technological products such as computers, phones, etc which have significantly improved in quality but no compensating increase in price. Particular cases exist where tremendous improvements in quality have actually coincided with price reductions.

2.10 Real vs. nominal – price indices

More often than not, you hear of economic growth. When the Minister for Finance presents the national budget, there is always a report of how much the economy has grown. This is precisely how our output has grown. If the growth is say 20%, it means this year's output is 20 percent above previous year's. The value of what we produce has increased. This is good because it tells that we now have more food, we have more of this and that because we produced more. This may sound very simple but two things we have to differentiate.

Assume a simple economy with only one producer of sugar. In year one, 20 packs are produced and the price is K11.50. The total value of output is the product of output and price. In this case, the economy will have its GDP as $GDP = 230$. In the year that follow, the total amount of sugar produced increases to 22 packs. Production has increased by two units. We know that prices are also not constant. They are bound to increase. Assume now the price has increased to K13.00. it means the total value of output in the second year is $GDP = 286$. Given this information, what is the percentage increase in GDP?

Given the two values, we would say GDP has grown by 24.3 percent. The value of output has gone up by that margin. But has output really gone up by that? Well, no. Initial output was 20 and the second is 22. This is clearly a 10 percent increase. We get two different increases because the former fails to take care of the changing price. The increase in value is over stated because the unit of measurement, the price, has also changed. If we control for this by assuming that prices remain unchanged, then you are talking of **real GDP** increase. When GDP is measured at market prices, that is varying prices, then it is called **nominal GDP**. Both measures have merits and demerits and the choice will more be guided by the motive of calculating GDP.

Activity 2-3

Mention circumstances under which nominal GDP is preferred to real GDP and vice versa.

The main difference between the two measures of GDP is that while nominal GDP ignores price changes, the real GDP assumes that prices remain constant. The latter more specifically measure the real part of GDP, the actual output. It removes the effect of changing prices on GDP. Then how are the two measures related?

If you are given the nominal GDP increase, you should by now, know that it includes increases in prices. This increase in prices is captured by what we refer to as **price index**. The price index measures how prices have cumulatively changed from the base year. The base year is simply the year in which you start to measure the price changes. Its selection is really arbitrary but may be influenced by certain events that significantly affect prices. For instance, Zambia rebased the currency on the first day of 2013. Such a year can then be taken as a base year because it is when a new set of pricing is introduced.

The price index, for each will then be a comparison of the prices in the review year with prices in 2013. Precisely, it is the ratio by which prices have increased (or decreased). It differs from inflation which is more concerned with change in prices from one period to the next. The introduction to inflation will be covered in depth in Unit 6.

The GDP, measured differently, has many uses. As a synonym to standard of living, GDP for different years can be used to assess the country's performance from one year to another. This will indicate whether there is improvement or deterioration in terms of wealth creation. In addition, GDP for various countries provides a better way of comparing the economic performance of different countries. You can tell, based on GDP, that one country is better than another. In comparing across countries, care must be taken to ensure that the GDP is measured or converted to a uniform currency. This can be done by way of exchange rates.

2.11 Determinants of national income; income and expenditure model

You have by now seen how income flows between various sectors of the economy. You have also learnt precisely how it can be measured. What is yet to be covered is „what determines national income?“ This subsection is therefore devoted to discussing the determinants of national income. We are looking for the level of income for which the economy is in equilibrium. Owners of factors of production receive incomes from which they spend on consumption. The other part that is saved obviously finds itself in investment as explained earlier. This expenditure will again generate incomes. This cycle of earning and spending goes on infinitely. At the end of the period, we are interested in the level of income.

A model has been devised to determine the equilibrium level of income. It is the income and expenditure model. Let us start with the Aggregate Expenditure equation given earlier

$$AD = C + I + G + X - M$$

the items on the right side of the equation are determined as follows. Consumption is a function of income. It will increase as income increases. Thus $C = a + c(Y - T)$. The a denotes autonomous part of consumption while c is the marginal propensity to consume, that is, a portion of every kwacha of disposable income that goes for consumption. Investment is independent of income but will be influenced by the interest rate. The discussion of interest rate is beyond the scope of this module. As such, investment expenditure will be treated as an exogenous variable.

The treatment of government expenditure will depend on assumptions we put across regarding the budget position and taxation. Though government expenditure is supported by the tax, the two need not be equal always. To keep the analysis simply however, we assume that government runs a balanced budget. This is a budget where expenditure equals taxations. With regard to taxation, we have to make an assumption of the tax system in operation. We restrict ourselves to basically two tax systems, the lump sum taxation and the proportional tax.

The trade balance is assumed to be exogenous to the model. Given this information, we can now put all the information together. We know that at equilibrium, the level of income will equal expenditure. Thus.

$$Y = AD$$

$$Y = C + I + G + NX$$

$$Y = a + c(Y - T) + I + G + NX$$

In the simple model where taxation is lump sum, this will simplify to (after making Y subject of formula)

$$Y^* = \frac{1}{1 - c} (a + I - cT + G + NX)$$

the assumption of a balanced budget will further simplify the equilibrium income to

$$Y^* = \frac{1}{1 - c} (a + I + (1 - c)G + NX)$$

Activity 2-4

Determine the equilibrium level of income if tax was a proportion of income, that is, $T = tY$.

In the model, the right side of the equation has the autonomous expenditure. This is multiplied by what is known as the income multiplier. It is a multiplier because it multiplies the change in an autonomous variable to get the change in income. To make it simple, assume that for whatever reason, autonomous expenditure increases by K1000. The income will go up by more than K1000. The reason is simple. The person that gets the first income as a result of the change in autonomous expenditure will also pass on the benefit in form of induced expenditure.

On average, the model predicts that income will go up by $\frac{1}{1-c}$ times the change in autonomous expenditure.

Activity 2-5

Given the following model of a closed economy (with no exports and imports)

$$AD = C + I + G$$

$$C = 70 + 0.8Y_d$$

$$I = 50,$$

$$G = 100$$

$$T = 0.25Y$$

What is the equilibrium level of income?

Find the budget surplus ($T - G$) at the equilibrium level of income

Redo b above when government exogenously increases expenditure to K200? K300?

2.12 Unit Summary

This unit, which is an introduction to National Income Accounts, discussed the ways in which income flows, which provides the stages at which it can be captured and measured. It has also brought out factors that determine national income. With this knowledge at the back of your mind, you will have no problem progressing to the next unit.

UNIT 3

3 UNEMPLOYMENT

3.1 Introduction

In political campaigns, or developmental gatherings, the issue of unemployment is usually central. There is always a talk of high levels of unemployment, high youth unemployment and so on. Often, the discussion will hinge on creating jobs for the unemployed. This already breaks the first nightmare. That is, the unemployed are those without jobs. But does this include the retired? the minors? And those simply not willing to be employed? For sure these are with no jobs as amply concluded.

This chapter is devoted to the topic of unemployment. Its primary concern is to discuss the meaning of unemployment and its ramifications on other macroeconomic fundamentals. In particular, it will define unemployment. It will also include a look at various types of unemployment which are of course linked to the causes. The unit will then look at the effects of unemployment. The unit will end with a discussion of cures for unemployment

3.2 Contents list

- Definition
- Types and Causes
- Effects
- Cure

3.3 Statement of pre – requisite knowledge

The unit does not assume any pre-knowledge on economics. The reader is nonetheless expected to have understood the first unit.

3.4 Learning objectives for the unit

By the end of the unit, the reader should be able to:

- Demonstrate the understanding of unemployment
- Identify the different types of unemployment
- Identify the different sources of unemployment
- Discuss the remedy for different types of unemployment
- Outline consequences of unemployment

3.5 Supplementary Reading

Begg, D., S. Fischer and R. Dornbusch (2003) Economics. (7th Edition). London: McGraw-Hill

Hardwick, P., B. Khan and J. Langmead (2006) Introduction to Modern Economics. London: Longman

Dornbusch R and S. Fischer (1993) Macroeconomics

Mankiw N.G. (2003) Macroeconomics

3.6 Time Required

The unit must take no more than two weeks.

3.7 Definition

In its lateral meaning, the word employed refers to someone in employment, someone with a job. The type or nature of the job here is peripheral. All that matters is that a person has a job. But the converse will not necessarily mean the opposite. That is, the unemployed are not necessarily those without a job. Lack of a job is but only one of the necessary requisites for being unemployed.

To get to the definition, let us start with some classification of the population not engaged in any employment along two variables. Simply asks everyone to identify themselves with the following two questions. Then summarise the results in a table below.

Table 3.1. Willingness and Ability to Work

		Are you willing to work for a
--	--	-------------------------------

		wage?	
		YES	NO
Are you able to work for a wage?	YES	A	B
	NO	C	D

The two classification basically hinge on ability and willingness to work. Those that will report ability are ones with the requisite body ability to work. The opposite will be those unable to work due to age (too young or too old), legal restrictions such as minimum age for one to get employed. These cannot be said to be unemployed because they do not possess what is required for them to work. The logic is simple. Even if there were plenty of job, would they work? The answer is NO. What prevents them from working is not the absence of job but their inability.

What about willingness? For various reasons, some people may simply not be willing to work. It could be they have acquired enough wealth and see no more reason to continue working. This applies well with those that go on early retirement. They still possess the ability to work but voluntarily decide not to. Another group is those that decide to go for school or higher learning. They too have the ability but have decided to do other things. Some people may also come out of employment in order to take care of children. This is more common among women. All these are not willing to engage in work in a particular period of time

In Table 3.1, region A represents the population that is both willing and able to work. This is the category referred to as the labour force. It is the total number of people that are both willing and able to work. Note that any one working is doing so voluntarily. This means all the employed are part of the labour force because we know they are willing and able to work from the fact that they are working. This conclusion only fails in the face of forced labour.

The other regions, B; C; and D, people are not working not because there are no jobs. They are either unwilling or unable to do so. These are said to be out of the labour force. They are not part of those that government must be concerned with. They have voluntarily decided to be out of employment. Even though government created an infinite number of jobs, *ceteris paribus*, these will still not be working.

To narrow to the point, the unemployed are those in A but have no jobs. They are in the labour force, able and willing to work, but there are not jobs available for them. They are the unemployed and the group of interest to government. Formally, a person is unemployed if they are willing and able to work but cannot find that job. The first part of the definition defines the labour force and the latter partitions the labour force into the employed and unemployed. Thus,

$$\text{Labour Force} = \text{Employed} + \text{Unemployed}$$

if we divide both sides by the labour force, two statistics emerge, the employment rate and the unemployment rate respectively.

$$100\% = \frac{\text{Employed}}{\text{Labour Force}} + \frac{\text{Unemployed}}{\text{Labour Force}}$$

the sum of the employment and unemployment rate always equal 100 percent. The implication is that the same inference can be drawn from either measure. An economy is said to be doing well if it has a higher employment rate or a low unemployment rate.

3.8 Types and Causes of Unemployment

Having defined what unemployment means, this subsection discusses the various types of unemployment. These are discussed below.

According to Mankiw (2003) frictional unemployment results because it takes time for workers to search and find new jobs that best suit their tastes and skills. If Mr. Banda has been working for Nakambala sugar in Mazabuka and for some reason decides to quit and find a new and probably better job, chances are that he will not find the new job there and then. He therefore becomes unemployed. His quitting does not entail going out of the labour force because both the principles of willingness and ability are still there. Between leaving a job and finding another one, Mr. Banda is said to be frictionally unemployed.

This type of unemployment will exist even at full employment, when the number of jobs equals the number of workers. The unemployment will exist because the workers and jobs will need to be paired. It will take time for Mr. Banda to know that the job he is after is in Kabwe, in the central part of Zambia. During this period, Mr.

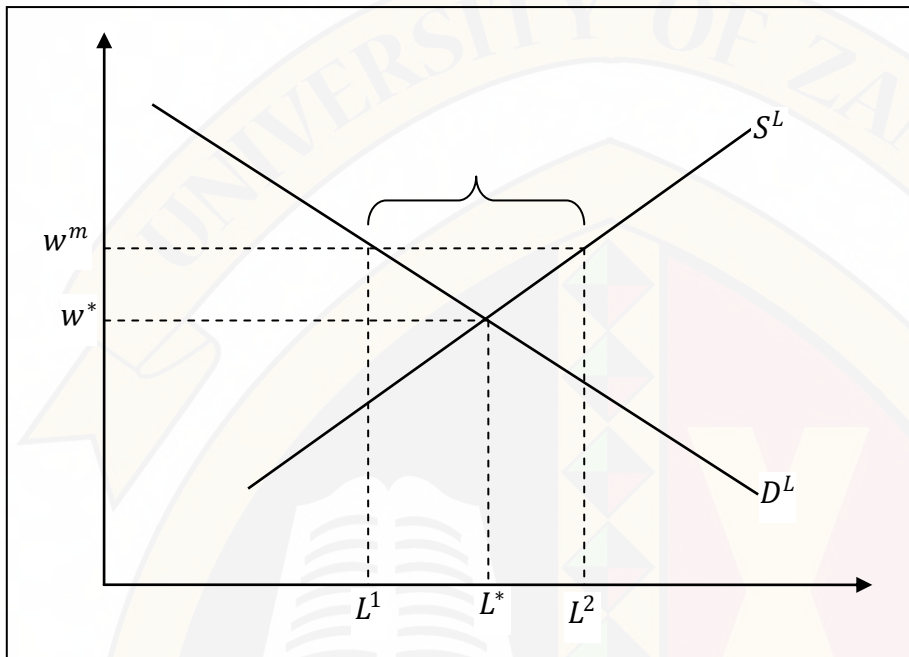
Banda will be said to be unemployed even though there is a vacancy waiting for him. Friction unemployment also includes fresh labour market entrants such as new graduates. Vacancies to take them may be available but they will need time and information to find them.

The other type and perhaps more strong is when there are simply no available jobs at the prevailing wage rate. People will be unemployed because the available jobs do not match the number of job seekers. There are more seekers than are available jobs. This is referred to as classical or real wage unemployment. It occurs when the supply of labour (employees) exceed the demand for labour (employers). Recall that in a free market with a flexible price (wage), demand and supply will always equal. But why doesn't this happen in the labour market? There are good reasons for this anomaly.

The first is that labour is not transacted like tomatoes on the market which has one price today and another one tomorrow. Labour markets are characterised by long term engagements or contracts. Once an employee and a company sign a contract, the wage rate will not be adjusted downwards simply because there is another fellow willing to work at a lower rate. The employer will continue with an expensive employee because they are more secure with the current one or the contract cannot be terminated without huge cost.

The second reason is that the market clearing wage rate may not be desirable in the eyes of the government. In such cases, a more desirable above equilibrium wage rate is set. It is called minimum wage rate. This is illustrated in the figure below.

Figure 3.1. Structural Unemployment



As depicted in Figure 3.1 above, w^* is the market clearing rate. If the market was allowed to operate at this rate, everyone would be employed. But how? Perhaps this is the most crucial issue when defining unemployment. The mechanism to full employment is simple. Some unemployed ($L^2 - L^*$) would go out of the labour force in frustration. They cannot work at the lower rate. Firms on the other hand will hire a few more workers ($L^* - L^1$) because labour has become cheaper. Adding the two will give the number of the initially unemployed. This however will not happen because government will have decreed that w^m is the lowest legally permissible wage rate. Long term relationships between employer and employee will also work along similar lines.

Seasonal variations in the demand for labour may also create some unemployment. In the agriculture sector for instance, there are peak and off-peak periods. During the peak periods, there is a surge in labour demand which pushes the wage rate upward. During the off peak period, the wage rate may fail to proportionately adjust downwards due to pressure from the employed. This will sustain an above-equilibrium wage rate resulting in unemployment. This may also be referred to as seasonal unemployment. The name derives from the fact that its occurrence is seasonal.

Seasonal fluctuations are always known to occur or complete within a year. But there are also long term fluctuations in economic activities which may well impact on employment levels. These are cyclical fluctuations and unemployment resulting from such is known as cyclical unemployment. During a boom, the demand for labour is high and so unemployment will be at its lowest. During recessions however, such as the financial crisis of 1998, the demand for labour is low creating or leading to high unemployment levels.

Three sources of unemployment have been discussed so far. We mentioned the frictional in which time must elapse before someone is landed with another job. Second was the above-equilibrium wage issue which creates excess labour. The third is due to seasonal/Cyclical fluctuations. An additional cause of unemployment may result from structural changes in the economy.

Consider the Zambian economy in the late 1990s when most mines were shutting down. A lot of labour was skilled and only skilled to work on the mines. When the mines were shutting down, other opportunities arose such as the agriculture and tourism. The problem however is that the labour under question was specialised in mining. It could not be retrained with ease so as to be absorbed in the other booming sector. In the short run, the labour is rigid and can only be employed in the mines. An insurmountable time is required to retrain this labour so that it is useful in other sector. This is referred to as structural unemployment. It comes about due to structural changes in the economy.

Structural unemployment can also be caused by technological advancement. As new technologies are introduced, they tend to displace the unskilled labour while creating opportunities for the skilled labour. For instance, mechanisation in the agriculture has rendered most unskilled labour redundant while creating opportunities for skilled machine operators. A person previously hired to weed on a farm will no longer be required. The farm will nonetheless still require services of someone to spray herbicides. Since the person previously hired for weeding lacks this skill, they will remain unemployed while a vacancy remains at their doorstep.

It must be clear from the foregoing that structural unemployment does not entail the absence of jobs. It is the mismatch between required skills and what is available. It is unemployment amidst vacancies.

3.9 Effects of Unemployment

The direct effect of unemployment is the lack of income which itself is the precursor to poverty. At the individual level, unemployment is responsible for the high poverty levels. Because people are unemployed, they will earn nothing. As a result, they have no adequate resources, at least in the long run, to meet even the basic needs.

At national level, unemployment of labour has the same effect as the unemployment of other resources. Recall that production is on the PPF only when all resources are employed efficiently. Since labour is one of the many resource endowments a country has, its underemployment or unemployment will prevent the economy from an optimal production path. There is room for accelerated growth if some labour remains unemployed.

In addition, unemployment has been observed to be behind the engagement of the youth into vices that are detrimental to the wellbeing of the nation. It is partly responsible for crime and uprisings in extreme cases.

3.10 Cures of Unemployment

How is unemployment cured? Well, this will depend on the cause or type. Basically four causes have been mentioned above. These are summarised in Table 3.2 below.

Table 3.2: Cause and Type of Unemployment

Cause	Type of Unemployment
Imperfect information	Frictional unemployment.
Too high wage rate	Real Wage or Classical unemployment.
Fluctuations in economic activities	Seasonal and Cyclical Unemployment.
Skill mismatch	Structural unemployment.

In order to come up with the cure, it is important to look at the causes. This sounds an easy task, but it may not be. The first problem of imperfect information can be dealt with government investment in information technology. Empower the labour market by improving mechanism through which job seekers find jobs or labour

seekers find employees. This is a market facilitation strategy. The government helps the market to be more efficient in its role of linking suppliers and demander of a particular good in this case labour.

The classical unemployment problem often puts government in a dilemma. It is also behind the much heated debate on issues of minimum wages. The minimum wage is meant to make jobs more decent by assuring the workers of a more reasonable pay. On the other hand, it creates excess labour supply, which we call unemployment. Thus, classical unemployment is actually caused by distortions in the labour market. These distortions are created or perpetuated by government's desire for better wages as well and the activities of the labour movement.

The best option for an economy on this type of unemployment is a political issue. It cannot be objectively decided because it depends on the weight put on the gain in welfare because of good wages and the loss arising from loss of jobs. We leave it to the reader to judge whether un-unionised labour markets with no government intervention would make the situation better than it is currently.

The other option to classical unemployment is to deal with the demand side for labour. Labour, as an input in production, is demanded because of its contribution to a marketable good. The government would enhance job creation by promoting investment (into especially labour intensive sectors). The increased investment will give birth to firms that will demand for more labour. This implies an outward shift in the demand for labour. More labour is employed at the same wage rate initially perceived too high.

Seasonal and cyclical unemployment is caused by fluctuations in economic activities in the economy. It can be managed by counter cyclical fiscal and monetary policies. These are policies that will promote job creation during recess and suppress economic activities during booms. The suppression of booms is particularly important because of the link between unemployment and inflation. Periods of low unemployment are often characterised by high inflation level. This is summarised in the famous Philip curve. This is also highlighted in the Unit on Inflation.

The skill mismatch problem causing structural unemployment is facilitated by enhancing labour retraining programmes. This type of unemployment arises because

the labour that was productive in the collapsing sectors fails to be absorbed by the emerging sectors because of the rigidity in skills. When the trouble of retraining is lessened, labour will easily become mobile across sectors. This is due to the ease with which labourers can acquire skill relevant to the new sectors.

Activity 3-1

The mining sector has been booming in Zambia until the early 2000 when most of them started to shut down citing the increasing cost and low copper prices. A lot of workers laid-off, Zambia was almost going on its knees. Discuss some of the steps government took to avert the situation. What else could they have done?

3.11 Unit Summary

Unemployment is a topical issue world over but more so in developing economies such as Zambia. Formulating a policy prescription to tackle it requires a good understanding of the causes because each remedial measure will only work for a particular cause of unemployment. After going through this module, it is now expected that the reader has improved their understanding of what unemployment is, what causes it and the different steps that can be taken to address it.

UNIT 4

4 GOVERNMENT SECTOR

4.1 Introduction

In the circular flow of income, three sectors were identified; the households, firms and government. The households are the owners of means of production while the firm does the actual productions. The output is then sold to households who pay from the factor payments received. The coming of government into the model did not change anything. Government is depicted as collecting taxes from both the firms and households. In return, government provides services which benefit both the firms and households. For example, roads are opened and maintained by the government, a big part of education, health etc are covered by the government.

This unit is devoted to discussing the government sector. This is a macro subject which is dealt with in a holistic manner. To do this, two key items will be discussed; that is taxation and government expenditure. Under taxation, the idea is not to consider micro elements of taxes. Rather, the objective remains to discuss the taxes in a macro manner. This will involve understanding what a tax is. We will also look at the nature or characteristics of a tax. This will also include a discussion of desirable taxation.

In the similar, manner, the government expenditure discussion will not be centred on individual elements or goods that government provides. Instead, it will be centred on the level of whole government expenditure. Since government spends from its tax collections, this unit will also make an exposition on balance of taxes over expenditure. Does the government need to spend only what it collects at any time? This will lead to a discussion of funding options for the government. This will be discussed under Fiscal Policy.

4.2 Contents list

- Public Finance
- Budget including Debt Management

- Fiscal Policy

4.3 Statement of pre – requisite knowledge

In this unit, it is expected that the student has read and understood Unit 2, on national Income Accounts. In particular, the student is expected to have grasped the connection of government to the two others.

4.4 Learning objectives for the unit

By the end of the unit, the reader should be able to:

- Show understanding of taxation
- Demonstrate understanding of Fiscal Policy
- Outline and explain fiscal policy instruments.

4.5 Supplementary Reading

Langmead, P et al. Tax Policy issues in Zambia: Selected Papers, Lusaka

4.6 Time required

This unit is expected to be covered in no more than two weeks.

4.7 Public Finance

This subunit is defined by two words which make it title; public and finance. Even for a novice, this must be clear enough that the subject under consideration has to do with finance. Indeed this is true. But, what kind of finance? Well, finance for the public. The finance belonging to the public essentially means government finance.

Almost all of government finance is taxation from the firms and households. This however does not rule out possible scenarios where government may raise substantial revenue from public corporations. A case in mind is Saudi Arabia. This country knows no tax. Government is able to get all the required finances from the oil industries it runs. In Zambia and many other lowly developed countries however, the opposite is true. While government does run some corporation such as ZESCO,

ZCCM-IH, NATSAVE, etc, these seldom remit any profits to government. They in most cases turn to government for recapitalization. For this reason, this unit will discuss public finance only from taxation perspective.

4.7.1 Taxes

In legal terms, a tax is defined as an involuntary charge on individuals and business by the state. This means taxes are monies that are paid to the government. Note from the definition that there is not attachment of a good or service in exchange. This differentiates taxes from other monies that may be paid to government in direct exchanges for a good. A tax is paid without a direct benefit to the payer. Because of this, the payer would under normal circumstance not be willing to pay. He/she pays because it is mandatory.

Taxes are broadly divided into two. The direct and indirect taxes. The dividing line is very thin and there is generally some disagreements from one jurisdiction to another. A direct tax is a tax collected directly when income is earned. Examples include the Personal Income Tax. This tax is charged whenever an income is earned. It is charged on wages, rent, interest or any other incomes earned. In Zambia, it is famously known as the Pay As You Earn (PAYE). Another example of direct tax is a corporate tax. This is charged on profits made by business entities.

Indirect taxes on the other hand tend to be indirectly charged. It is normally charged when we spend. The notable example is the Value Added Tax (VAT). You will not see VAT on you pay slip. But every time you walk into a shop, you pay it indirectly. It is paid through goods and services that we buy in everyday life.

Activity 4-1

Explain what a Customs and Excise duty are

Are they direct or indirect taxation?

The question of direct or indirect tax cannot be resolved solely by looking at the mode of payment. For in both case, payment can be in any form, cash cheque or transfer. Strictly speaking, a tax is direct if the charge is made directly on the payer. This is easily understood if tax incidence is considered first.

Take a simple example. A shop owner is required to pay a sales tax at 10% of the price. Assume his goods were initially K20.00 each. Given a 10% tax, the shop owner has two choices. Maintain the price and pay K4.00 to government. This effectively reduces the price due to the seller. The second option is to pass the tax to the buyers by increasing the price. Assume the seller is able to push the price to K25.00. After paying the 10% to government, the seller remains with K20.00 per unit. Though the tax was charged on the shop owner, it is clear from the above example that it can actually be shifted to another. This is what defines an indirect tax. To refine the earlier definition, a direct tax is one whose burden cannot be transferred. It is normally imposed on persons or property. Indirect taxes tend to be imposed on transactions which the affected are at liberty to avoid. They are also transferable as seen in the above example.

Activity 4-2

List and explain three indirect taxes not mentioned in the above paragraph

4.7.2 Retrogressive, Proportional and Progressive Taxes

Almost all contemporary taxes are not independent on the payer's income. In the past, there was what was referred to as Poll tax. This was levied on an individual irrespective of their income status. This was viewed as the most unfair tax because it treated the *unequal* equally. Taxes that are based on income level, for individual or businesses can be classified into three; retrogressive, proportion and progressive taxes.

A tax is retrogressive if the percentage of tax to income reduces with an increase in income. In percentage terms, the poor pay more than the rich. Under this definition, a poll tax qualifies as a retrospective tax. If the amounts under the poll tax are converted into percentages of income, the proportion would decline as income goes up. What the poor pay would be a big proportion of their income while for the rich; it might actually be a negligible portion.

A proportional tax is one where everyone pays a fix portion of their income. If it is 30% corporate tax, every business will have to pay only 30% of their income. This is

the most common type of corporate tax. They all pay a fixed proportion of their income irrespective of the level. The third is the progressive tax. A progressive tax system is one in which the percentage increases with income. The rich do not only pay high amounts; they also pay a high proportion of their incomes in tax. This is the system currently used in Zambia's income tax.

4.8 Budget including Debt Management

What really is a budget? The term budget is used in all circles, at individual level, household, community or at state level. Every now and then, we talk of budget. At the beginning of every month or year, a family will sit down to make a budget. This will consist of items or goods that they plan to buy or acquire.

Activity 4-3

Have you ever been involved in making a budget?

If Yes, outline what you were budgeting for and how you went about doing it.

Basically a budget is a plan on revenue and expenditure for an entity. At household level, the key question is often what to buy. This is the expenditure side of the budget. The revenue side is rarely an issue for one simple reason. As individuals, we lack control of our revenue (salaries). We do not decide how much to have in a particular month or year. For this reason, we take them as given. Once the revenue part is known, a family or individual goes on to formulate the expenditure side.

For a government however, they have control of both the expenditure as well as the revenue. They can decide how much to collect (procedurally of course) by employing various taxations mentioned earlier in the unit. A government budget will therefore outline how much government plans to collect and how that is going to be collected. On the expenditure side, it will give the various goods and services that government will spend on.

4.8.1 *Balanced Budget and Debt Management*

For every budgeting entity, the possibility of saving or borrowing is not ruled out. For a household, it is possible to spend more than is earned by resorting to borrowing. But borrowing simply means consuming now and paying later. The borrowed money will ultimately have to be paid back. The same applies for savings. They are not meant to last for eternity. They must be consumed at some point. So revenue and expenditure may be unequal in some periods but must be equal over lifetime.

In a similar manner, government revenues and expenditures need to be equal at all times. We define a budget deficit as the gap between expenditure and revenue.

$$BD = G - T$$

The acronym BD represents budget deficit, G stands for government purchases of goods and services and T is tax revenues. This model assumes no transfer payments. When there is neither deficit nor surplus, that is $BD = 0$, the budget is said to be balanced. It is often called a balanced budget. It is balanced because the revenue and expenditure sides are balanced. They are equal and government spends exactly what it collects.

Since there is room for debt (or saving which is rare), government can decide to spend more than it receives by borrowing. Government borrows by issuing Treasury Bills and Bonds which are short and long term loans to government respectively. When $G > T$, we say there is a budget deficit. This must not be confused with an overrun which is unintentional. A deficit is part of the plan and will include plans on how it is to be covered. The opposite is called a budget surplus. Some money is saved and available for loaning to deficit states. But what is the limit to borrowing?

Government can borrow as much as it desires provided the source is there. As long as there are entities willing to lend to government, it will borrow. The moral limit however is that the borrowed money will ultimately have to be paid back. Thus borrowing must be limited by the ability to pay back. So how much should government collect in tax and how much must be borrowed? Well, such a question can be ably answered by turning to fiscal policy. This is discussed in the succeeding subunit.

4.9 Fiscal Policy

Government expenditure on goods and services adds to the aggregate demand. The more government spends, the higher the aggregate demand. The tax does not directly appear in the model. It nonetheless affect aggregate demand through consumption and Investment. When tax is high, net income is low and so will consumption. For firms, taxation will eat into net profit (Profit after Tax) which results in less money available for reinvestment or dividend payment. Thus both G and T are important components of aggregate demand.

This provides government with an ability to manage aggregate demand. As reported in Wonnacott and Wonnacott (1982), government has the ability – and the responsibility- to manage aggregate demand to ensure continuous prosperity. This is achieved through a fiscal policy. This is a policy on government expenditure and taxation. During periods of high inflation caused by too much aggregate demand, government can restrain aggregate demand through high taxation or reduced government expenditure. In periods of high unemployment caused by insufficient aggregate demand, government can increase its expenditure or cut taxes in order to boost aggregate demand.

Using fiscal policy, government can reduce fluctuations in aggregate demand so as to ensure the economy grows smoothly. When fiscal policy is used to expand aggregate demand, it is known as expansionary fiscal policy. it is intended to increase aggregate demand usually in times of depressions. When aggregate demand is too high, this will manifest in high inflation as will be seen in Unit 6. In such case, the ideal steps are to restrain or contract aggregate demand. This is referred to as contractionary fiscal policy.

4.10 Unit Summary

The role of government in an economy cannot be underscored. As a dominant player in most economies, actions of government do have far reaching consequences on other players. In addition, government activities can also be used to give direction for economic activities in the nation. This unit has endeavored to help the reader understand what government does in the economy, how it manages its finances. It is expected that the reader now understands well how government expenditure and

different types of tax system can be used to steer the economy in the desired direction, that is, through fiscal policy.



UNIT 5

5 MONETARY SECTOR

5.1 Introduction

In every economy, money has become a critical medium through which transactions are conducted. Everyone everywhere uses money to buy various goods and services. This unit is devoted to discussing the monetary sector of the economy. In particular, the unit will look at money from a macro point of view. That is, the importance of money will be looked at a national level as opposed to considering individual decisions on money.

The unit will commence by asking the question, „what is money?“ We all know what money is but answering this question is no near easy. A reader is expected to devote good amount of time to understand money from an economical view. The unit will also look at the banking system of the country. This will include a discussion of the central bank and its role in the sector. A discussion of monetary policy will close the unit.

5.2 Contents list

- What is money?
- The banking system and the central bank
- Monetary policy

5.3 Statement of pre – requisite knowledge

The unit does not require any prerequisite knowledge

5.4 Learning objectives for the unit

By the end of the unit, the reader should be able to

- Demonstrate understanding of money
- Outline how banks operate in an economy

- Formulate monetary policies

5.5 Supplementary Reading

Dornbusch, R. Et al (2004) Macroeconomics 9th ed.

5.6 Time required

The unit should take no more than two weeks

5.7 Money

Virtually every country has a sovereign right to have its own currency or what we call money. Names may be common but each will be identifiable to a specific country. For instance, many countries use the denonym „dollar“. The currency name is always extended so as to specify the country. The same applies to the „Kwacha“. It is used in Zambia as well as in Malawi. But this does not in any way mean the two countries are using the same currency. The names are extended to *Zambian Kwacha* and *Malawian Kwacha* respectively.

Given the above background, how then is money defined? Well, it still remains difficult. To sum it all, money lack a concise and scientific definition. It is generally understood as anything that is generally accepted in exchange (Dornbusch et al, 2004). It is a material holding no value in itself. It is accepted in payment only because of the belief that it will later also be accepted in payment by others (Ibid). The actual material used for money has evolved from metals such as gold and silver to mere paper and electronic entries in banks.

It is nonetheless defined based on its functions and characteristics. We turn to look at the functions of money and then the desirable characteristics.

5.7.1 Functions of Money

Think of what money does in your life, church, community. It basically serves three functions. These are the basic functions which some writers break into many specific functions.

Transaction: Money is used in many transactions. Whenever one is buying, there is an exchange of money. You give away K5.00 in exchange for K5.00 worth of airtime. The buyer receives the commodity while the seller receives money.

Store of value: In addition to buying things, some value may need to be stored for future use. For instance, a farmer will harvest his/her output around May. The output must be spread until next harvest twelve months later. This means storing some output. With poor storage facilities and the nature of the commodity, this may not always be possible. An alternative is to sell everything and store the money. That means money functions as a store of value.

Unit of Account: if we are to compare the worthiness of two or more people, we have to look at their assets which obviously will be diverse. Some will report having cattle while others will have buildings. To be able to make a comparison, everything must be converted to a common unit of measurement. We convert all assets to monetary value. This means money has provided a unit we can use to measure something, wealth.

Money commodities have been used as money in the past. In the developed world, gold, silver, etc were the common commodities people used to store value and buy goods and service. In Africa and Zambia in particular, animals such as cattle were commonly used to settle debt. They remain a common mode of payment for „lobola“ to date.

Activity 5-1

Give two reasons why „cattle“ is not a good form of money.

5.7.2 Characteristics of Money

Money has been defined on the basis of its functions. The definition has however failed to provide for a closed meaning, a meaning devoid of misrepresentations or misinterpretation. This does also not pin point to what the reader already knows about money, *the notes and coins*. The definition recognizes anything that would

satisfy, albeit to varying degrees, the three stated functions; that is, a unit of account, for exchanges and store of value. Many commodities can be listed based on this definition.

Activity 5-2

Zimbabwe saw high levels of price increases, with essential commodities increasing two, threefold in a matter of hours. This later let to abandoning of the local currency in preference for the South African Rand and US Dollar.

Discuss the factors behind increasing prices

Why would a seller refuse to accept the Zimbabwe Dollar now?

To narrow the meaning of money in fundamental sense, the desirable characteristics of money are used to rule out some possibilities. Note that these are *desirable* characteristics. They need not exist always. In a situation with many possible commodities for money, the desirable characteristics provide criteria for picking the best commodity. The characteristics are:

Acceptability: Money must be acceptable by the general populace if it is to function as a medium of exchange. The seller must have confidence in the money they are receiving in exchange for the commodity. Recall that money does not have an intrinsic value. Its value is extrinsic. It is based on the commodity it can buy. From this, we can state as follows: „Money buys because it is acceptable and it is acceptable because it can buy“.

Portable and Easy to Carry: money is used in transactions, both planned and unplanned. A transaction is planned if it is anticipated. It is unplanned if unanticipated. Individuals will often carry money *in case* In an event a lucrative opportunity to buy something, I must have ready cash or I lose the chance. This works well with portable money. Imagine of carrying bags of maize on our errands for precautionary purpose. This is made easy with weightless paper money. The advent of Automated Teller Machines (ATM) has even taken this desirable quality further.

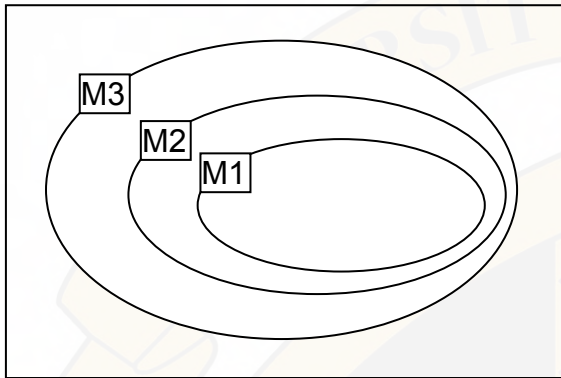
Divisibility: One of the roles of money is transaction, that is, for buying goods and services. This will range from high value goods like automobiles to low value goods like *chewing gums*. In order for money to be used in all transactions, it must be divisible enough to be able to buy the very lowest commodity. It should have smaller denominations so that is always possible to divide it into any reasonable values. This means cattle are not good money. It can't be used to buy a loaf of bread.

Relative Scarcity: Who ever that has money possess some power. They have the purchasing power from which they can command any good of equivalent value. Like any power, purchasing power must be sparingly available. It should be accessed through means of offering a good or a service. This will make man and woman work solely to have some purchasing power. In the process, each contributes to the welfare of society through production. If however money is as plentiful as sands of the shore, no man would work for there would be no incentive in work. It is therefore a desirable characteristic that money is relatively scarce.

Durable: the other role played by money is the store of value. A person must able to store some value in money form. If the commodity used for money is perishable or for whatever reason cannot be kept for a long time, then this purpose is defeated. The commodity itself cannot be kept for long, it would be worse the value it possesses.

Based on how each type of money meets the desirable characteristics, we can assign each type to a category. Three categories of money are identified which depend mainly on the liquidity level of money. That is, the ease of converting the said money into other assets or goods. The diagrammatical presentation of the categories is shown in Figure 5.1.

Figure 5.1: Categories of Money



As shown in the Venn diagram above (Figure 5.1) money is in three categories. The categories are coded M1, M2 and M3. We discuss these below.

M1. The most common function of money, one that all of us have known money for, is that of medium of exchange. M1 is basically money based on this function. This is a restrictive definition of money as it relies on one function only. In Zambia, this will include notes and coins issued by the central bank. It will also include checking deposits. Notes and coins are used in exchange in our daily lives and it should not be surprising that they form M1. The demand deposits on the other hand may not be very common, but not as rare as the term „cheque“. A cheque is a payment order issued on an account. The account holder issues cheques as an alternative to paying by cash. Cheques enable account holders to use money held in checkable deposits much as they would use notes and coins. It must be clear however that it is balances in accounts that count as money and not a written cheque. The same applies to Debit cards. They are not money but merely provide a means of transferring money. The buyer gets a commodity and money is transferred by the bank through an internal mechanism. To summaries, M1 is notes and coins in circulation by demand deposits held with commercial banks.

M2 is a little broader than M1. It consists of M1 and other short and medium term deposits with commercial banks. To understand this definition, the reader will need to carefully understand the difference between demand deposits and short/medium term deposits. The former are deposits solely for safe keeping by the bank. They have no time attached and the depositor can access the money at any time. The latter on the other hand are binding deposits. The depositor commits to maintaining

the money for a given period of time, usually ranging from a month to just under a year. Money held in short or medium term accounts may not be accessible at any time. The depositor can nonetheless access the money in a reasonable time.

M2 is a broader measure of money supply in an economy. It is also the widely used measure of money supply. It is however still not the comprehensive measure. We turn to M3.

M3 is the addition of M2 and long term deposits. It is the most comprehensive definition of money as it takes into account the three functions on money. In summary, M1 is a subset of M2, it is contained in M2. The same applies for M2 and M3. Each measure is broadening on the previous.

5.8 The banking system and the central bank

By now, the reader is very conversant with banks. Most probably, you have dealt with a bank before or actually have a *savings* account. The banks, which have now increased in number, are part of a system called the banking system. This subunit is devoted to discussing the banking system. Where necessary, examples will in as far as possible be Zambian. This is in order to bring out the subject matter with a local flavour.

5.8.1 The Banking System.

In every system, there are always players and regulators. The Zambian banking system is no exception. It comprises of the Central Bank and Commercial banks and other non bank financial institutions. The central bank is the regulator. Its detailed discussion is deferred to section 5.8.2. The main players in the banking system are commercial banks such as Barclays, Finance Bank, ZANACO Bank and many others. Their number has grown since independence to just below 20 as of the year 2013². Their role is to provide different financial services to the public. They take deposits from which they disburse loans. The number has grown since inception to just below 20 as of the year 2013.

² www.boz.zm

Commercial banks also facilitate electronic payment system. When you buy a Cooker from Game by cheque, there is no physical money transfer involved. Assuming both the buyer and the seller bank with the same bank, only a change of figures will be done. The bank will credit one account and debit the other. In this case, the payment will have been facilitated by the bank. The same applies when Debit cards are used. The bank will effect the payment in the background.

In addition to Commercial Banks which are deposit taking institutions, the banking system also has the non-bank financial institutions. As the name may suggest, these are not banks but deal in finances. They include Leasing Companies, Building societies, micro financial institutions and Bureau de changes. These are specialized institution which do not take deposits but are able to advance loans.

The loans will vary in size and often for specific purposes as in the case of Building societies. Micro financial institutions deal with micro credit to small and medium enterprises including households. Examples include Bayport Financial Services, Micro Bankers Trust, Blue Financial Services Zambia Ltd and many more. In Zambia, micro financial institutions and bureau de changes dominate the non bank financial institutions. The non-bank financial institutions often deal with micro credit. They borrow huge loans from banks; repackage them into small loans mostly targeted at the households and small enterprises.

5.8.2 The Central Bank

In every monetary economy, there is always the central bank. It is called by different names from one economy to another but its role is almost common. In Zambia, the Bank of Zambia is the central bank. It performs a number of key functions which include the regulation of the banking system of the country. Like in many countries, Bank of Zambia is publicly owned institution whose main objective is not profit maximization. Rather, its aim is to “formulate and implement monetary and supervisory policies that ensure the maintenance of price and financial system stability so as to promote balanced macro-economic development”.

As an institution established by an act of Parliament, the Bank of Zambia is charged with the following specific responsibilities³.

- Licence, supervise and regulate the activities of banks and financial institutions so as to promote the safe, sound and efficient operations and development of the financial system.
- Promote efficient payment mechanisms
- Issue notes and coin to be legal tender in the republic and regulate all matters relating to currency
- Act as banker and fiscal agent to the republic
- Support the efficient operation of the exchange system
- Act as advisor to government on matters relating to economic and monetary management.

This will be discussed in the subsection that follows.

5.9 Monetary policy

Just like every commodity, money too has a supply and demand side. The interaction of the two ultimately determines the market price of money, interest rate. The demand side of money is made up of the need to hold real balances. It is defined as the amount of (liquid) real money that people plan to hold. People often have the desire and actually hold some liquid cash. The reasons vary from one person to another but can broadly be described as transaction, precautionary, and speculative. These motives relate the three broad functions of money discussed earlier.

The **transaction demand** for money is the need to hold money so as to remain liquid. We want to have the needed cash so that we are able to buy the commodities we intend to buy. For instance, let's take two people, a farmer who only comes to the bank once in a week and an urban dweller. The farmer can only access a bank once in a week while the latter has access during the normal operating hours. We emphasise on operating hours because we want to hold that banks are not accessible throughout.

³ The Bank of Zambia (Amendment) Act, 1998

When deciding on how much money to withdraw, the farmer will have to figure all the payments he is likely to make in the whole week. For the urban person, one week will be too much. He will only think of a day since he can easily get to the bank should there be need for more liquidity.

This is known as the transaction demand for money because it is driven by the need to be able to make transactions. It is positively related to one's income because the richer one is, the more (value) transaction they plan to make. While a rich person may plan to buy durable items, the poor fellow is limited by the income and can only plan to petty items.

The transaction demand for money relates to planned expenditure. These are items that we plan and even write down. With a high level of uncertainty, households will also hold money for unforeseen eventualities. If driving, a breakdown is not planned but must be expected. There are also chances a family member may get ill. All these reasons make people hold liquid cash beyond what they plan to spend. This is called precautionary demand for money. It is similar to the transaction demand except that it is meant for unforeseen expenditure. Like the former, it will also depend on the level of income. The higher the income, the more expenditure people will instantaneously want to make. They will thus carry more money for „just in case“.

The third motivation for holding money relates to the function of money as a „store of value“. Households keep value in many commodities such as animals, grain etc. An asset is suitable for storage if it does not lose value. Grains will therefore not be suitable because by being perishable, it cannot be stored for long without losing value. Keeping animals for value sake may also have its own advantages and disadvantages. Moreover, there are cultural preferences to the commodity for storing.

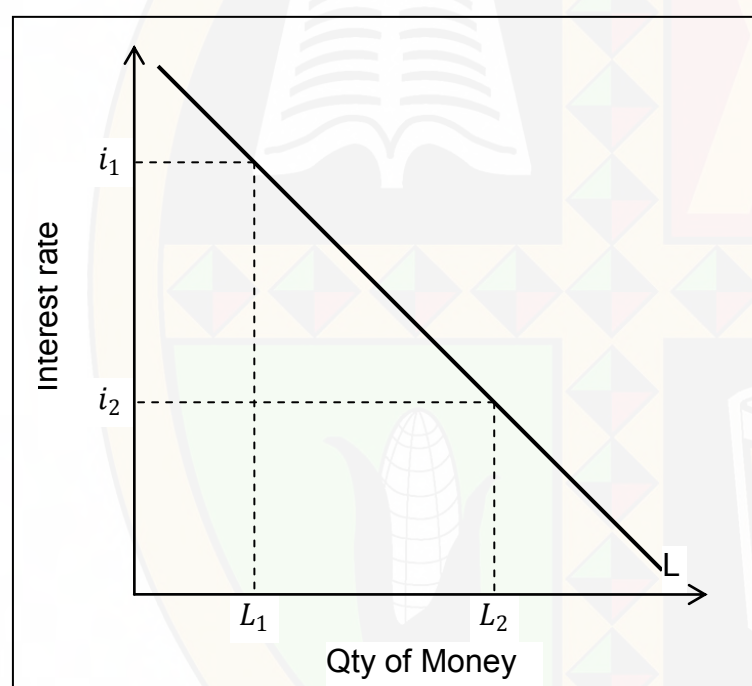
A culturally neutral commodity is money. Simply keep cash. It has an added advantage because it can be converted to other assets with ease. This reason for holding money is known as speculative demand for money. Money is held mainly on a speculative basis on unforeseen investment opportunities. But money can be kept in two places; at home or in the bank. Since the demand for money is concerned with money held outside the banking system, we hold that this function will fall with

increased incentives offered by banks. If bank offer high interest for the deposit, it will act as an opportunity cost of holding money. As such, households will demand or hold less for speculative purposes.

With lower interest rate, there is little to lose by keeping liquid cash. Therefore, the quantity of money demanded will increase as interest rate falls.

Many other factors beyond the scope of this module will influence the demand for money. The demand function will look like presented in Figure 5.2: The demand for money below. As can be seen, it is no different from the demand for any other commodity you may have encountered already.

Figure 5.2: The demand for money



Like any other good, money has a price; that is interest rate. It will act like the price of bread which will depend on the demand and supply of the commodity. Equally for money, the demand and supply side will determine the interest rate.

As a macro price, the interest rate is of keen interest to monetary authorities as it affects investment. The interest rate is the price of money (or loan) and when it is high, the cost of investing borrowed funds also increases. Authorities will want a certain level of interest rate depending on the prevailing circumstances and overall economic goals. Ordinarily, the monetary authority will alter the interest rate by

playing with the supply side of money. When money supply is increased, (*-ceteris paribus-*) the interest rate falls. The same applies for a reduction in money supply. This leads to the definition of monetary policy.

Simply put, monetary policy is the management of the availability of money so as to affect the interest rate with an ultimate objective of price stability and stable economic growth. At high interest rate which inhibits investment, the monetary authority or the central bank can increase money supply. This will lower the interest rate and in turn boost investment. In the extreme case, this may overstretch the economy forcing inflation⁴ higher. The authority may again need to apply some „slow-down“ mechanism. Reduce money supply so that the interest rate goes up. this will push out some investment leading to reduced aggregate demand⁵. With reduced demand, prices will not rise any faster.

So how does the central bank regulate the amount of money in circulation? Will it come on streets and dish out cash in the name of monetary policy? Or confiscate money because they want to reduce money supply? Absolutely not. There are systematic ways of achieving reduced or increased money supply. They are called instruments of monetary policy.

5.9.1 Instruments of Monetary Policy

The instruments of monetary policy are tools that the central bank can use to alter the availability of money. They include instruments that will increase or reduce the stock of money as well as instruments that will deal with the complexity of accessing the money. The central bank will choose from among the many instrument on the guidance of effectiveness of the instruments. We discuss the instruments here. But a discussion of when different instruments are effective is beyond the scope of this module. This is covered at the intermediate level of theory.

Open Market Operations: this is the act of buying or selling securities such as bonds and treasury bills by the central bank. When the central bank buys securities, it pays by cash. This increases the stock of money in the banking system effectively

⁴ Inflation and how it is affected by monetary policy is discussed in the next Unit.

⁵ The aggregate demand model includes investment demand. With high interest rates, firms will not be able to borrow much for investment thereby leading to reduced investment. This leads to a fall in aggregate demand given by $AD = C + I + G + X - M$

increasing money supply. Conversely, the central bank may sell securities. This is done in order to reduce money stock in the system. Firms or individual buying securities will write cheques on balances held with commercial banks or pay cash. Either way, money is retrieved by the central bank. This reduces money supply through a multiplier effect.

The Discount Rate: You will recall that one of the functions of the central bank is lender of last resort. Commercial banks in need of money will often turn to the central bank. The central bank will lend at a rate specially called *the discount rate*. It is the interest rate charged by the central bank to commercial banks that borrow from it to meet temporary needs for reserves⁶. The central bank will use the discount rate to remotely control the lending behaviour of commercial banks. When the rate is high, commercial banks will be more sceptical on lending and will keep high reserves to be able to meet any demand. With a lower rate, commercial banks will be loose in lending because they can turn to the central banks at low cost should need arise.

The Policy rate: this is a guiding rate on the interest rates charged by commercial banks. The central bank will play with the discount rate to give a signal to the market on the direction of interest rate. It is not binding on the regulated banks but is *religiously* followed because of high level of competition in the banking sector and as a moral suasion matter from a supervising authority.

The reserve ratio: Commercial banks always need to keep a portion of deposits liquid. This is in order to meet its obligation of making available money whenever the depositor makes a demand of it. Banks are able to figure out the minimal proportion, just enough to meet daily obligation. They will keep the minimum since they must loan out as much as possible to maximise profits (interest receipts). But by doing this, a high risk is place on the bank of failing to meet its obligation. This is catastrophic and has potential to cause a *run*.

To avoid this, the central bank prescribes the minimum portions obviously above what banks would rationally hold. This is known as the required reserve ratio. Though this is meant to protect depositors, the central bank can take advantage and use it for monetary policy purpose. It will set a high ratio to suppress loan and reduce

⁶ Dornbusch, R. Et al (2004) Macroeconomics 9th ed, New York, McGraw Hill Co Inc.

money supply. Alternatively, it can prescribe a lower ratio if its goal is to promote increased investment.

5.10 Unit Summary

This unit has explained what money is and the role it plays in the economy. It has also discussed the banking system in the economy which describes how money is expanded or contracted. The reader is therefore expected to have improved understanding of the banking system in general. One must be able to describe the role of various players in the system such as the Central Bank, Commercial Banks and so on. Finally, the reader should be able to describe how a combination of the fiscal policy covered in the earlier unit and monetary policy can be used to drive economic development.



UNIT 6

6 INFLATION

6.1 Introduction

The measurement of economic performance through GDP is made complex by two factors. Firstly, we produce an array of goods; cattle, maize, copper, fish, electricity, honey, etc. These products cannot be added for they are measured in different units. Even if all measured in *kilograms* for instance, a „kilo“ each will have different value. To solve this problem, all products are measured in a common monetary value. Convert everything to *kwacha*. But this does not the whole problem.

The second factor is that the monetary scale is itself elastic. This poses a problem when comparing GDP across time. The increase in prices will inflate the measured GDP. In order to control for this, it is a requirement that we keep track of how prices have changed so that we can expunge its effect on GDP through deflating. The changes in price occupy a special room in economic theory and this unit is devoted to discussing this. It discusses inflation.

6.2 Contents list

- Definitions
- Causes
- Effects
- Cures

6.3 Statement of pre – requisite knowledge

No pre-requisite knowledge is assumed in this unit.

6.4 Learning objectives for the unit

By the end of this unit, the reader is expected to:

- Demonstrate understanding of inflation

- Be able to explain the meaning of changes in inflation level.
- Be able to prescribe remedies for inflation

6.5 Supplementary Reading

Begg, D., S. Fischer and R. Dornbusch (2003) Economics. (7th Edition). London: McGraw-Hill

Hardwick, P., B. Khan and J. Langmead (2006) Introduction to Modern Economics. London: Longman

Dornbusch R and S. Fischer (1993) Macroeconomics

Mankiw N.G. (2003) Macroeconomics

6.6 Time required

The reader will need utmost one week to fully comprehend this unit.

6.7 Definitions

Inflation is defined in many ways. Some will simply describe it as representing the loss of value for money caused by more money chasing few goods. Besides, this is how inflation affects the ordinary person. It erodes the value of money because goods on average have become more expensive. Formally, inflation is the sustained increase in the general price level.

The definition brings out three key terms that deserve attention. First, inflation is not restricted to changes in the price of maize-meal alone, or fuel alone. It is concerned with the general price level. This means inflation is an average of how „all“ prices have increased. Second, the increase ought to be sustained. It must increase over time as opposed to one-time adjustment in prices. Third, inflation is positive when prices are increasing and negative when falling.

6.8 Causes

There are two schools of thought on the causes of inflation; the Keynesian and the monetarist view. The Keynesian look at inflation from a Demand and Supply point of

view, arguing that inflation is as a result of excess demand. The monetarists on the other hand view inflation as a purely monetary issue. We discuss the two lines of thought in detail below.

6.8.1 *The Monetarist View*

The quantity theory of money forms the basis for the monetarist view of inflation. We narrate it as follows. An economy produces goods and services. As indicated in the earlier units, these goods are only summed when converted to monetary units. Thus, total GDP is given as the quantity multiplied by the price level. Give multiple products, we sum the value of all the products, that is, $P_i Q_i$ where P and Q is price and quantity of a particular product i . The quantity theory of money simplifies this by simply giving

$$GDP = P * Q$$

The above equation represents the real side of the economy, the goods and services side. But for every good given away, there is an equal value of money received. Goods are sold or exchanged for money of equal value. This is also amplified in National Income Accounts which gives different but equal ways of measuring GDP. What we are trying to bring home is that the value of money in the economy must equal nominal GDP (GDP measured at market prices). We could then write $M = P * Q$.

M represents the total monetary value of all notes and coins in circulation. But are notes and coins used only once? Definitely not. The trader pays for transportation from the notes and coins received from customers. The bus driver will not get new notes to buy fuel but will use the same notes passengers paid for the services. In simple, notes and coins are used several times in a given period say year. This will modify the above equation to

$$M * V = P * Q$$

this is known as the „equation of exchange“. The V is the velocity of money or the number of times money is used in transactions. This is the perceived average. For instance, if a K100 note changes hands ten times in a year. It means it has been used to buy goods worth K1, 000.

Now take logs of the function and differentiate. Since this is differentiation of logs, the result is interpreted as percentage changes or simply growth. The equation will be in the form $m + v = \pi + g$ where

- m is the growth in money supply
- v is the growth in the velocity of money
- π is the growth in prices which we have already defined as inflation
- g is the growth in real output. This is also called real growth rate or growth in actual output.

The velocity of money is assumed to be constant. Therefore, its growth rate v is zero. After making this adjustment to the equation and making inflation π the subject of formula, we get

$$\pi = m - g$$

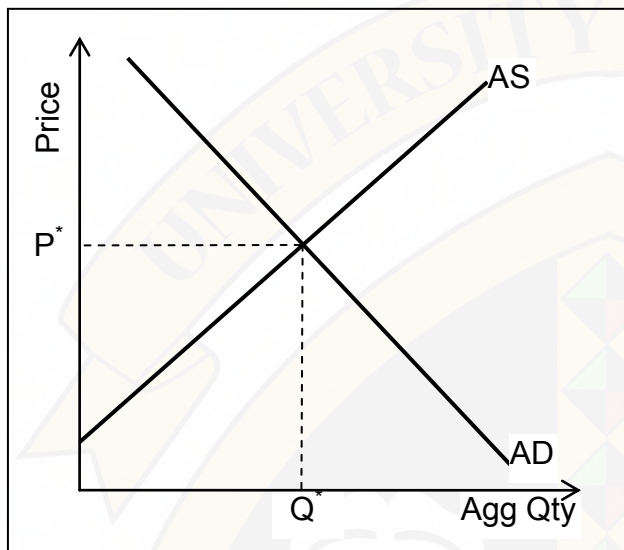
this equation will define inflation as the excess money supply growth over real output growth. This is exactly how inflation is popularly known, as a purely monetary phenomenon. If money supply grows at the same rate as real output, inflation would be zero. It's simple. The additional money only comes to go with with the additional output. But when money growth is faster, there is some additional money with no matching additional output. Prices will go up and we call it inflation.

It is important to understand this equation carefully. It does not say inflation is solely a result of increasing money supply. Instead, emphasis must be placed on the difference in growth rates. The implication is that, with a low real growth rate, even a moderate growth in money supply will feed into inflation. When the real growth rate is high however, there is greater scope to grow money supply without worrying about inflation.

6.8.2 The Keynesian

For the Keynesian, a market demand and supply is the starting point. We could look at the market for cattle, maize, copper and so on. But remember inflation is a general price issue not single price. Therefore, we deal the aggregate market. We must look at the entire economy market. Like any other market, there is the aggregate demand and the aggregate supply.

Figure 6.1: Aggregate demand and supply



While the demand curve is always assumed to slope downwards, there are three possible scenarios for the supply curve. These pertain to the level of employment in the economy. When the economy is highly underemployed, the supply curve will be horizontal. It is possible to increase quantity without necessarily affecting the general price. Towards full employment, the supply curve slopes upwards. It means output can be increased but will come with increased price.

When everything is employed, output is at its maximum. It is no longer possible to increase output, *ceteris paribus*. The response of the price to changes in aggregate demand will depend on the shape of the aggregate supply function. We assume for this analysis that the aggregate demand slopes downwards.

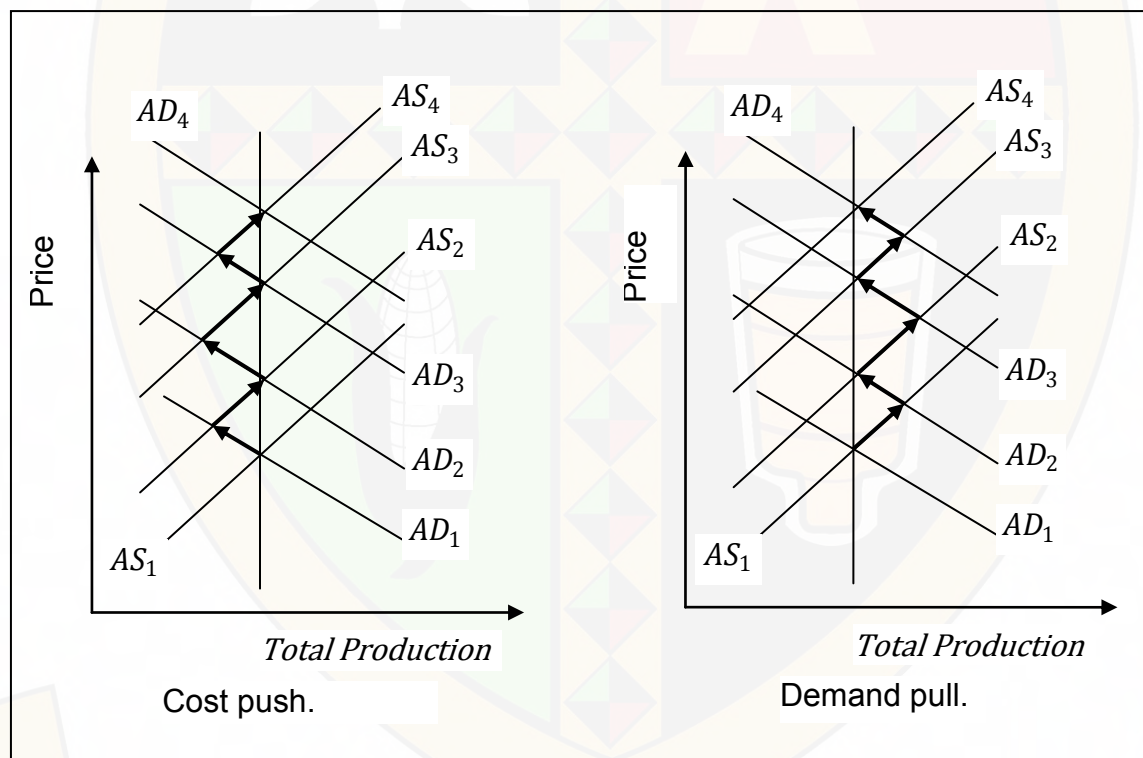
One of the two sides of the market, demand and supply, will trigger changes in the price. When the demand moves first, prices will go up. Workers will realize a loss of value of their earnings and will demand for higher wages. When firms concede to this, they pass on the increased cost of production to consumers through further increase in price. Again the worker will push for wage increases. This will form a cycle of wage-price increases. This is called demand pull inflation. It is demand pull only because it is triggered by the demand side.

When prices increases are triggered by worker asking for higher wages, it is called cost push inflation. Again, the name is solely based on the side that sets the wage price spiral in motion. Once the spiral is on, it is not possible to differentiate the two

types of inflation, demand or cost triggered. When producers are justifying increases in commodity prices, they will often cite increases in the cost of production. We all know that this is as a result of awarded salary increments. But it would be politically unsound for the firm to say so. The fact any way is that cost has gone up because we have awarded worker higher salaries.

When it is the workers' turn, they will site increases in commodity prices such as mealie meal, transport and so on. ZCTU or its affiliates will always talk of increased cost of living. But what constitutes the cost of living? It is food, transport and accommodation. The producers of food will have increased prices because their workers (-who are part of ZCTU-) were awarded salary increment. The same applies to transport providers; they have to deal with drivers. Figure 6.2 below shows the demand pull and cost push inflation.

Figure 6.2: Demand pull and Cost push inflation



The noticeable differences from the above diagram is that cost push inflation results in under employment. Firms react to increased factor cost by employing less and as a result producing less. On the other hand, demand pull inflation results in over stretching of the economy. At full employment, there is additional demand which can

only be met by short term adjustments. Factor cost will go up because they are overstretched. The two causes set in motion a wage-price spiral which has a lot of inertia. It often requires politically unsound decisions to put it to a halt.

6.8.3 Imported Inflation

In addition to the above causes of inflation, there is also a part of inflation which is imported. Zambia depends on imports for oil. If the price of oil on the international market goes up, the effect will also carry over. It will lead to increased production cost for all producers using oil. Producers will realise that they are not making as much profits as before. They will increase the output price in a bit to maintain the profit margin. An increase in commodity prices will erode the real value of wages. Thus workers will also follow and demand for higher wages. The wage-price spiral will have been set in motion. This is called imported inflation. The same applies for countries like the Congo DR which depend on imported electricity. If producers of electricity hike tariff, the importing countries will import at an inflated price. This will be imported inflation for the importing country. In the exporting country, the cause could be demand pull or cost push.

Irrespective of the cause, inflation is the same. It has the same effects as we shall see below.

6.9 Effects of Inflation

To discuss the effects of inflation, we must differentiate between two categories of inflation. There is an anticipated inflation and unanticipated inflation. The former is expected and taken into consideration when making future contracts. The latter however is never expected and is not hedged.

6.9.1 Anticipated Inflation

When inflation is perfectly anticipated, the market players take necessary steps to safeguard their value. When engaging in future transaction such as debt, the lender will ensure that the repayment takes into account the inflation level. They know that at the time of repayment, prices will have gone up which erodes the value of money. Therefore, an inflation premium is added to ensure the maintenance of the real value.

Under such circumstances, inflation does not have real cost to the economy. Nonetheless, there are two sources of, abet intrinsic, cost of inflation even when perfectly anticipated. The first is referred to as the *shoe leather cost*. For money held in banks, the interest payment takes care of inflation. But no interest is paid on money held in cash. There is no indexing for cash money and it therefore loses value with higher inflation levels. To lessen this effect, wealth holders or receivers of money will make numerous journeys to banks just to ensure that every kwacha received is banked as quickly as possible. The cost of such manoeuvres may be negligible. But when made so frequently, the cost becomes quite tangible. The shoe leather cost therefore refers to the cost of time and energy that is spent to counter the effect of inflation on money. The advent of online banking will reshape this view to inflation.

The second source of cost is called the *menu cost*. In every shop you go into, even as small as a barber shop, there is a price list. In big shops using computerised systems, the price list also needs to be configured in the system. When inflation is low, prices will not change so often. With frequent price changes however, the price list will have to be changed at the same rate. There may also be need for services of IT specialist to reconfigure the system in view of new prices. This tends to be costly and an inconvenience when done so often. This is known as the menu cost because it involves changes to the menu, physically or electronically.

6.9.2 Unanticipated Inflation

Inflation is unanticipated if it occurs unexpectedly. Of course we always expect inflation to be there. The difference is in the level of inflation. For instance, there may be reputable predictions that inflation would be 8 percent. If it actually turns out to be 8 percent, we say it is perfectly anticipated. If however, actual inflation is say 12 percent, the 4 percent excess is what constitutes unanticipated inflation. Thus, unanticipated inflation is the excess over what was anticipated. This is expressed in the equation below.

$$\pi = \pi^a + \pi^u$$

where π^a is the anticipated inflation and π^u is unanticipated.

The total inflation is the sum of anticipated and unanticipated inflation. The former is taken care of as discussed in the preceding subsection. For the latter, it is simply unanticipated. No body expects it. As a result, economic agents will not take any precautionary measure against its effects. The following are the main effect on inflation when it is unanticipated.

Risk of future transactions. Given two people entering into a future payment agreement, one is the receiver and the other is the payer. The payment will take into account the price level at the time of payment. They look at the anticipated inflation and fix the amount. When actual inflation turns out to be higher, the receiver loses out since the prices have gone up more than anticipated. The amount received is over eroded. When inflation is lower than anticipated (negative anticipated inflation) the payer will pay more in real terms. The anticipated inflation will work against the payer.

In either case, the presence of unanticipated inflation brings into play the notion of risk in future transactions. There is risk because both the payer and receiver have non-zero probability of losing out in the transaction. Economic agents will respond to this risk by reducing their activities in future contracts. The ultimate effect is a reduction in lending activities which are critical for economic development.

Wealth Redistribution: this effect is related to the effect presented above. Because of unanticipated inflation, there tends to be some redistribution of real income from fixed income earners to flexible income earners. For instance, pensioners earn monthly payments that were fixed years back. Because they are not flexible (it cannot be increased in view of increased commodity prices), the accumulated effects of inflation have rendered the payments valueless.

Inflation also redistributes real value from regulated sectors to unregulated sectors. We can use the example of the energy sector whose prices are always subject to approval and a retail sector that can adjust prices at will. When it becomes expedient to increase prices because of increasing cost of production, the retail sector will increase prices and maintain its profit margin even when the cost of production has gone up. For the regulated sector however, the prices cannot be altered before a lengthy approval process. During this process, the firm loses out on its profit

because the cost side has gone up without a compensating adjustment on the revenue side.

6.10 Cures for Inflation

Inflation is a sophisticated issue in economics. Economists seldom agree on the course of action to deal with inflation. The disagreement starts with the question of whether inflation is really something that must be dealt with. If it is to be dealt with, what are the best actions to take? This lack of consensus emanates from the strong link between the level of inflation and unemployment. There is a trade-off between inflation and unemployment. This is depicted by the famous Philip curve.

To reduce unemployment, expansionary policies are often pursued. This may involve increased budget deficit or increased money growth. In both case, there is pressure exerted on inflation. On the other hand, reducing inflation requires use of contractionary policies which hinder employment creation. It is this trade-off which leaves economist in limbo. One's position will depend on the seriousness they attach to the problem of unemployment. Economists that place a higher premium on unemployment will often have a liberal approach to inflation.

Regardless of one's view on inflation, there exist remedies to inflation. Again, the remedies will depend on the cause of inflation. We first deal with the monetarist view on inflation. Recall from the earlier subunit that, according to this school of thought, inflation is caused by excess money growth over the real growth rate. Therefore, the remedies or cure lies on the central bank to control money growth. The monetarists conclude that inflation will be minimal if money supply growth rate is kept closer to real output growth.

Under the Keynesian, there is the demand pull and cost push inflation. It was stated that demand pull is as a result of excess demand, obviously caused by increases in aggregate demand not matched with supply. Cost push inflation is caused by a fall in supply often coming as a result of increased cost of production.

Dealing with demand pull inflation is much easier. It is caused by excessive demand. To control this, take measures that restrict demand. We say this is easy because it has no negative side effects. Since demand exceeded supply at full employment, a

contractionary policy, monetary or fiscal, will simply bring demand to full employment. The possible measures include fiscal policy which can be used to restrict consumption through taxation or more directly, reducing government purchase of goods and services. Alternatively, a contractionary monetary policy can be used. This will increase interest rates which ultimately reduce investment expenditure.

Cost push inflation is a little complex. Cost push results from increased production cost. It comes with it underemployment which is responsible for the reduction in output. The reader must pick it very clearly that cost push inflation is attached to unemployment. That is, it presents two problems, inflation and unemployment. Taking measures prescribed for demand pull will succeed in bringing inflation down. But in addition, the measures are also likely to reinforce the problem of unemployment.

In such cases, it is very difficult for the government to decide on the course of action. Government has to weigh the benefits of reduced or cured inflation against the sacrifice that often has to be made in terms of unemployment. This is the reason we said dealing with cost push inflation is more complex than the former. It has to be approached from the supply side by dealing with output. The government can take measures that address hindrances to increased production such as low investment. This view is also supported by the monetarist group which views inflation as a gap between money growth and real growth rate ($\pi = m - g$). The country can maintain a high growth rate in money supply but reduce inflation through enhanced real growth rate.

What if inflation is imported? Often, imported inflation will manifest as cost push inflation. Massively imported commodities are rarely final goods but inputs in production of final goods. When it comes out as cost push inflation, the remedies discussed above should be applied. This is the case with importation of crude oil. When its price on the international market goes up, it becomes cost push in Zambia. The solution would be to cushion the effect of an increase in oil price on general output in the country. Promotion of investment that will see output buoyant can be considered. Production or input subsidy can also be considered although with additional problems.

Lastly, a more *laissez faire* action to inflation is to do nothing. We have defined inflation as a measure of the speed of prices. But why do prices rise? It is because of imbalances in the market. Prices go up in order to ensure the market clears. An increase in prices is often towards clearing the market. Therefore, inflation would theoretically cease when the supply catches up with the demand.

Moreover, not all inflation is bad. Because of the trade-off between inflation and unemployment highlighted in the Philips curve, so amount of inflation may be favourable in order to promote the growth of the economy and reduce unemployment. Wages tend to be sticky downwards. There would be unrest if any firm or government proposes a wage reduction. Inflation therefore provides a surreptitious way of reducing wages so that firms can hire more labour. Workers often take time to realise that their real wages have gone down because inflation tends to be silent.

6.11 Unit Summary

Generally, an average person knows that an increase in inflation is an indication that things are not as they ought to be. However, the reduction in the level of inflation is often misunderstood to imply that prices will soon become bearable, that is, they will fall. We expect that the reader is no longer at that stage. After going through the unit, the reader is also expected to fully appreciate the meaning of inflation and its interlink with other macroeconomic variables such as unemployment and monetary policy, both covered in earlier units. The unit has discussed the different causes and corresponding cures of inflation. The reader should therefore be able to define effective policy measures which must be based on the identified cause.

UNIT 7

7 EXTERNAL SECTOR

7.1 Introduction

In a global economy, every nation depends on the others for many goods and services. In Zambia, we produce a lot of copper which has no market domestically, (at least to match production). Its production is driven by its demand from outside. We also consume many commodities whose production process, we hold the least idea. We are able to use computers, consume common salt. But these commodities are not produced anywhere in Zambia. They are bought from outside or simply imported.

Much as we have looked at what can be done domestically by the economic players, there is also need to look at the performance of the outside economy because it does exert some influence on the performance of the domestic economy. Our ability to export largely depends on the happenings on the international market. It depends on the ability and willingness of other countries to absorb our output. This section is devoted to discuss the external sector.

7.2 Contents list

- Trade
- Balance of Payment
- Exchange Rates

7.3 Statement of pre – requisite knowledge

The unit requires that the reader has a firm understanding of the circular flow of income. This is presented in Unit 1.

7.4 Learning objectives for the unit

By the end of this unit, the reader is expected to be able to

- Demonstrate understanding of the case for international trade
- Discuss the different components of the Balance of Payment
- Demonstrate understanding of exchanges rates

7.5 Supplementary Reading

Dornbusch, R. Et al (2004) Macroeconomics 9th ed.

7.6 Time required

This Unit is envisaged to take no more than two (2) weeks.

7.7 Trade

The large part of microeconomics is anchored on production and consumption. This was highlighted in the circular flow of income. Firms produce by using factors hired from the households. The households on the other hand consume the output from firms. Trade between firms and households ensures that output goes to where it is demanded and factors of production are accessed by firms. In the simplest terms, trade is the process through which goods and services are sold and bought.

When trade takes place across the international boundaries, it is called international trade. The selling of goods and services to other countries is called export while buying from other countries is known as importation. Thus exports denoted by X are the total monetary value of all goods and services that the country sells to the international market. By exporting, we receive payment in international currency such as US Dollars or South African Rand.

A larger volume of exports is an indication of good performance of the economy. The economy receives foreign currency from exports. Jobs are also created in order to boost production in view of the *perfectly elastic* international demand.

Imports on the other hand are denoted by M and represent the total monetary value of all goods and services the country buys from outside. It is immaterial here to look at the actual buyer. Whether this trade is initiated or conducted by government, firms or households, does not alter how international trade affects the economy.

In the aggregate demand model, the export and import appear as an addition and subtraction from the model respectively. The equation is $AD = C + I + G + X - M$

But dealings on the international market are not restricted to trade. There are other transaction which do not fit into the definition of trade but still result in movement of money across the border. These include foreign investment, international transfer payments. Foreign investment is now a common terminology in Zambia. We have mines almost entirely run by foreigners. We also have major retail outlets such as Shoprite, Spar and many others. These are owned by outsiders who have invested in them. At the time of investment, the foreigners brought money into the country, for investment.

When profits are declared, they must be expatriated since the owners are outside the country. Thus dividend payments will often lead to money going out of the country. In addition, when business is wound up, any proceeds from liquidation have to be sent out of the country to wherever the owners are. All these transactions do not fall in the ambits of trade. They do not involve an exchange of money for a good.

7.8 A Justification for Trade

International trade often involves haulage of goods over long distances. No doubt this increases the cost of providing the goods in the destination markets. If every country is able to produce all it needs (at differing costs though), why engage in trade? Two principles have been developed that explain why every country fights to participate in international trade. These are *absolute advantage* and *comparative advantage* principles. The two principles are covered in detail in higher level economics courses. We only introduce them here.

7.8.1 Absolute Advantage

This principle was developed by Adam Smith. Assume two countries, A and B producing two goods, food and electronics. Each country is free to commit all resources to the production of one good only or a combination in any ratio. Because of different resource endowments such as size and quality of land and other factors affecting the production of the two goods, we expect the two countries to produce different quantities. For instance, Zambia has a larger arable land than Kuwait in the

Middle East. On the other hand, Kuwait enjoys massive crude oil reserves. As such, we expect Zambia to be able to produce more agriculture output while Kuwait will produce more oil.

Assume a constant Marginal Rate of Transformation production possibility schedules for the two countries are given as:

Country	If Food only	If Electronics only
A	60	20
B	30	30

If country A produced food only, it would produce 60 units. If all resources are used in electronics only, 20 units are produced. For country B, it produces 30 units of food or 30 units of electronics. Country A is better in food production since it can produce a quantity that B can never produce, *ceteris paribus*. We say country A has an absolute advantage in food production. Similarly, country B has absolute production in electronics production. The principle of absolute advantage postulates that countries will benefit from trade when each produces the good of its absolute advantage. We illustrate this below.

Take the above levels of production. The actual combination of the two goods will be determined by the preferences of each economy. Let us assume that each country will commit half its resources to each commodity. Then each will produce exactly half the above numbers as follows.

Country	If Food only	If Electronics only
A	30	10
B	15	15
Total	45	25

Since there is not trade, the above output is all that will be available for consumption. With the possibility of trade, assume now that each country produces only the product of its absolute advantage. Then we have the following table of outputs

Country	If Food only	If Electronics only
A	60	0
B	0	30
Total	60	30

Clearly, the total quantity has increased with specialization. With 60 units of food, country A can claim its initial level of 30 and B gets its 15. There is still 15 units left for the two to share. Similarly, there are now 30 units of electronics compared to 25 before specialization. This is a clear proof that countries with absolute advantage in different goods will mutually benefit by specializing and trading.

7.8.2 Comparative Advantage

The principle of Comparative advantage was also developed by Adam Smith but refined by David Ricardo. It argues that countries could still benefit from trade even if one country does not have absolute advantage in any product. While absolute advantage was based on ability to produce more, comparative advantage is based on lower opportunity cost. That is, a country has comparative advantage in one good if its opportunity cost (the forgone output of the other) is lowest. This often comes into play when one country has absolute advantage in both goods. Think of South Africa and Zambia. South Africa may probably be better than Zambia in the production of both food and electronics.

Let us have another PPF with a constant MRT as before.

Country	If Food only	If Electronics only
A	60	32
B	30	30

All the quantities have remained unchanged except than country A is now able to produce a maximum of 32 units of electronics. Now country A has absolute advantage in both goods. They may probably look down on B, that they cannot benefit anything by engaging in trade. Let us examine this case.

Again, since each country commits half of its resources to each product, country A will produce 30 units for food and 16 units of electronics. Country B will have 15 of each. Now let us identify the comparative advantage. We have said it is the opportunity cost, the loss in terms of the other product. If country A produces 60

units of food, it forgoes 32 units of electronics. That is, to produce one unity of food, country A sacrifices or give up $\frac{32}{60} = 0.533$ units of electronics. To produce a unity of electronics, A will forgo $\frac{60}{32} = 1.875$ units of food.

For country B, a unity of food, it has to sacrifice $\frac{30}{30} = 1$ unit of electronics. In fact, for country B the transformation is *one to one*. Clearly, it is much cheaper for country A to produce food. Therefore, country A has a comparative advantage in food production. Corollary, the cost of producing electronics is lower in B than it is in A. Country A has to give up 1.875 units of the food while B only has to part away with 1 unit of food. Thus, country B has a comparative advantage in the production of electronics.

Assume now that each country specializes in the product of comparative advantage, that is, country A produces food while country B produces electronics. Total outputs are shown in the schedule below.

Country	If Food only	If Electronics only
A	60	0
B	0	30
Total	60	30

The reader will notice that the quantity of electronics has fallen from 31 to exactly 30. One unit has been lost. To conceive the benefits to both countries, assume that country B only gets 15 units of each so that it is at the same level of welfare. That means country A will go with 45 units of food and 15 units of electronics. Country A has lost out on electronics (the blame is squarely on trade).

But how much compensation does country A get for the lost unity of electronics? They get 15 units of food. But their internal MRT only required getting 1.875 units. Clearly, country A has been over compensated. This means some units of food can actually be given to B and leave country A at a welfare level higher than before specialization. The units given to country B will make it move to a higher level of welfare. At the end, both countries has benefited from specialization and trade. We conclude that even without absolute advantage for each country, trade still offers mutual benefits to trading countries.

Activity 7-1

Assume a constant marginal rate of transformation for this problem. Suppose that Country A can produce a maximum of 56 textiles or 20 units of food. Country B is able to produce 50 textiles maximum or 10 units of food. Is there room for the two countries to benefit from trade? If so, how much should each produce? (Hint: Use comparative advantage principle).

7.9 Balance of Payment

We discussed the many ways of measuring the performance of the economy. These included the GDP and GNP. These are indicative of the economic performance of the country. Inflation is also an indicator of economic performance measured from the angle of price stability. In the same way, economist and policy makers are interested in knowing how the country is fairing on the external sector.

There are two measures used. The first measures the performance of the country in trading with others countries. It compares all trade revenues against trade expenditures across the boundaries. This is the Balance of Trade (BoT). It is an account of all transactions involving movement of money across the international border. The Balance of Payment for Zambia will show all monies coming into the country against monies going out of the country.

Dornbusch et al (2004) defines the Balance of Payment (BoP) as the record of the transactions of the residents of a country with the rest of the world. Since transactions are simply an exchange of money, the balance of payment will record all monies coming and going out of a country. Two main accounts divide the BoP. These are current and capital accounts. The BoP is in deficit if in net, the country pays out more than it receives. It will be in surplus when the opposite is true.

7.9.1 Current Account

This name must not be confused with the current assets and liabilities encountered in the Balance sheet of an institute. In the BoP, the current account records all international trade in goods and services. In addition, the current account will also

have a record of transfer payment, payments that are not a result of trade such as donations. The entity making and receiving the donation is immaterial here. Donations may be made by government or private individual. The same will apply on the receiving side. All that matters is that additional foreign money has come into the country or some local money has left the country.

In the case of Zambia, the current account will contain the value of copper as well as other non-traditional-exports on the credit side. In addition to export of merchandise, Zambia also exports services such as tourism. When international tourist come to view the many attractions Zambia is endowed with, they bring in foreign currency. When our Universities enrol international students, it is an export of services and will enter as a surplus entry in the current account. The current account will also include all aid (international) coming into the country. Zambia receives a lot of aid, as budget support to government as well as project funding. Most NGOs run on international donor funds. These will be part of the transfer payments adding to the current account.

On the debt side, all imports of goods and services plus donations that Zambia makes to other countries. The reader will probably have engaged in an international transaction before. When a Zambian is buying a used motor vehicle from Japan or designer shoes from Italy or coffee from South Africa, our foreign currency is sent out of the country. This will enter on the debt side of the accounts. They are deficit transactions.

Activity 7-2

Most mining companies in Zambia are owned by foreigners. When profits or dividends are declared, they must be sent to wherever the owners are, outside Zambia. Explain why this will fall under the current account

In addition, Zambia may import services such as local students pursuing studies in other countries. Take an example of Bwalya who enrolls at the University of Zimbabwe. Bwalya will have to pay for tuition in Zimbabwe, he will also have to pay for many incidental cost such as accommodation and food. But will the University of Zimbabwe accept Zambian kwacha? Will boarding house owners accept Zambian

kwacha? Well, the answer may seem ambiguous because of the special relationship between Zimbabwe and Zambia. But would it still be ambiguous for other countries like Ghana? Not really. Zambia's currency will generally not be accepted outside Zambia.

Given this situation, Bwalya will have to pay for everything using an internationally accepted currency. He will take dollars from Zambia into Zimbabwe. On the balance, Zambia will have lost on that value of dollars because it has imported a service in form of education.

Activity 7-3

Suppose Bwalya who enrolls at the University of Zimbabwe departs Zambia with US\$1000 for his expenses in Zimbabwe. Unfortunately, the total expenses in Zimbabwe amount to US\$1,500 and so he only pays for two thirds of its. Luckily, he is permitted to engage in part time works from where he earns exactly US\$500. How much should be recorded in Zambia's current account.

Zambia also sends money outside as transfer payments. Though it may not be the government making the donations, Zambian residents with relatives outside will often send money for various uses. The use again is not important for us. The concern of the current account is to record international movement of money.

7.9.2 Capital Account

While the current accounts records payments for goods or services, the capital account is a record of international investments. It is a record of purchases (and sales) of assets such as stocks, bonds, land, etc. It is a capital account because it deals with movement of capital across international boundary. The divide from the former may be difficult to see and care must be exercised. The current account is a record of transactions that are a result of currently produced goods or services as labour, capital or goods. It will not result in any future payments.

The capital account is a record of transactions intended to generate future payments. Because we are looking at the intention of generating future payments, capital goods

will basically deal with investment transactions. In Zambia, the phrase Foreign Direct Investment may not be common but its acronym FDI has become a vocabulary of every development Economist. In general, investment need not be direct only.

Not very long ago, Zambia sold its debut „euro bond“ on the international market to the tune of US\$750m. This money was not a payment for anything. It came into Zambia because the owners are hoping to get something as a result, that is, interest payment. This is one example of a capital account transaction, a purchase of assets.

There is also a lot of foreign investment in the mining of Zambia. All this represents inflow of foreign currency. The flow is investment in nature and is entered in the capital account.

Conversely, when Zambians invest in foreign assets such as land, capital, treasury bills or bonds, such flows are capital in nature. They enter the capital account as deficit transactions. This category will also include transaction such as Zambia buying land in Madagascar to build a resident mission there. It is all the same, some money has left Zambia. Again, we emphasise that it matter not as to who actually pays, government or private individuals.

To summarise, each account is in surplus if more is received than is given out. It will be in deficit if more has left compared to inflows. The summation of the two accounts should normally give zero. That means a deficit in the current account should be supported by a surplus in the capital account. The reasoning is simple. The foreign currency is not minted locally. So any payments outside (deficit transaction) must be sources from receipts from outside (surplus transaction). We can only pay from what we have received.

$$\text{current account} + \text{capital account} = 0$$

When the deficit in one account does not equal the surplus in the other, the above equation will fail. In that case, we say the country has a deficit or surplus balance of payments as the case may be. Specifically, a surplus occurs when the surplus account outweighs the deficit account. This means, on the overall, more has come in than has left. The effect on the economy is that there are now more foreign reserves

than before. The official reserves with the central bank will increase by the amount of the difference. The opposite will take place in the wake of a deficit.

7.10 Exchange Rates

From the name, it is much easy to construct the definition. Exchange rate implies the rate at which goods or services exchanges. In the barter system, it was known how much maize would exchange with a fully grown chicken. In economics, the term exchange rate is exclusively used in references to currencies or money. Therefore, an exchange rate is the rate at which two (and only two) currencies exchange. For instance, $US\$1 = ZMK5.30$ is one exchange rate. It shows that rate at which the dollar will exchange with a kwacha. A different rate will hold for each pair of currencies. For instance, we can have the rate at which the local currency exchanges with major international currencies.

In other words, the exchange rate is the price of one currency in terms of another. It is the rate at which one currency (say the US Dollar) will exchange with another, normally the local currency (Kwacha). Note that this exchange rate, like any other price, is for a specific market. That is, the dollar kwacha exchange rate in Lusaka need not be the same as in Livingstone for instance. In fact, the rate will even differ just across the corridor due to the imperfection of the market. Table 7.1 below shows exchange rates for four major currencies in Zambia. These are: the US Dollar, the British Pound, the Euro and the South African Rand.

Table 7.1: Exchange rates of four major trading currencies

Date	US Dollar			GB Pound			Euro			SA Rand		
	Buy	Sell	Mid	Buy	Sell	Mid	Buy	Sell	Mid	Buy	Sell	Mid
6 dec 2013	5.57	5.59	5.58	9.10	9.14	9.12	7.61	7.64	7.62	0.53	0.53	0.53
5 dec 2013	5.56	5.58	5.57	9.08	9.12	9.10	7.55	7.58	7.57	0.53	0.53	0.53
4 dec 2013	5.56	5.58	5.57	9.10	9.13	9.11	7.55	7.58	7.57	0.53	0.54	0.54
3 dec 2013	5.54	5.56	5.55	9.09	9.13	9.11	7.52	7.54	7.53	0.53	0.54	0.53
2 dec 2013	5.54	5.56	5.55	9.07	9.11	9.09	7.50	7.53	7.51	0.54	0.54	0.54

Source: Bank of Zambia

From the table above, you will see that roughly, a US dollar will sell for K5.57. The British pound will fetch about K9.11. The same rate of exchange can be used to

compare the worthiness of different currencies or wealth levels, assuming everything else remains unchanged.

Since currencies are sold on markets like another commodity, and its price responding to market fundamental, the dealers in foreign currency are motivated by the need to make profit. They will buy the foreign current from people wishing to get the kwacha and sell to those wishing to get the foreign currency. In order to make profit, they have to sell the currency at slightly above the purchase price. As such, a foreign exchange dealer will always display two rates; the buying rate and the selling rate. The former is the rate at which the dealer is buying a particular currency (foreign) while the latter is the rate at which he is selling. The latter will always be above the former.

Activity 7-4

Given the exchange rates in Table 7.1, suppose Mr. Lisulo is purchasing a Motor Vehicle from Japan invoiced at \$2,135 and the transaction is executed on 3rd December using a bank transfer. How much will be deducted (debited) from Mr. Lisulo's kwacha account?

All the rates discussed above pertain to the foreign currency against a local one. When the rate refers to two foreign currencies, it is called cross exchange rate. For instance, given the exchange rates to two currencies (in terms of Kwacha) we can also find how one foreign currency will exchange for another foreign currency. Simply divide the two exchange rates. From the table, to get the cross exchange rate for the Pound to Dollar

$$\frac{\$}{\pounds} = \frac{K/\pounds}{K/\$} = \frac{9.09}{5.55} = 1.64^7$$

The above cross exchange rate implies that a British Pound will exchange for US\$1.64 in Zambia. The rationale behind the formula is simple. Let us start with a person wishing to change her pounds to dollars. When the direct exchange is not available, he will have to first change the pounds to kwacha and then change the

⁷ Using the exchange rate of 2nd December, 2003 in Table 7.1

kwacha to dollars. Assuming no profit margins for the dealers, the resultant rate will be as derived above. In the same manner, we are able to evaluate the cross exchange rates for all the other currencies.

Activity 7-5

Refer to Activity 7-4. Suppose now that Mr. Lisulo decides to pay for his \$2,135 motor vehicle from his British Pound Account held with one of the banks. How much will be deducted from the account?

7.11 Unit Summary

The unit has discussed the basis upon which countries must trade. The reader is now able to understand the implication of trade in as much as it affects movements of money across the border. The reader should be able to link that with the concepts of international reserves which is often cited by political analyst as an indication of economic performance. The unit however does not go to look at trade policy, the reasons and nature of hindrance to free trade. These are covered in advanced economics courses. In conclusion, the unit has discussed the subject of exchange rates which the reader should be able to link to the demand and supply of foreign currency.

8 MODULE SUMMARY

The module is an introduction to macroeconomics theory. It provides a basis for the study of higher level macroeconomic courses at second, third and fourth year level. It has brought out six theme areas in macroeconomics, that is, National Income Accounts; Unemployment; the Government sector; the Monetary Sector; Inflation; and the External Sector. The reader is expected to fully appreciate these subjects in order to proceed to the higher level courses. One need to bear in mind though that this is an introductory course. As such, some technical requirements or subjects are left for succeeding courses.

